Spatial–Spectral Multiscale Mamba Fusion Network for Hyperspectral and LiDAR Data Classification

Bin Zhao, *Member, IEEE*, Wenjie Cao, Lin Han, Weiwei Song, Wenjuan Zhang, *Member, IEEE*, Lianru Gao, *Senior Member, IEEE*, Jakob Sigurdsson, *Member, IEEE*, Magnus O. Ulfarsson, *Senior Member, IEEE*, and Jón Atli Benediktsson, *Life Fellow, IEEE*

***Abstract*—Joint classification of multimodal remote sensing data, such as hyperspectral image (HSI) and light detection and ranging (LiDAR), is important for Earth observation. Early deep learning-based methods usually adopt a single network to extract the features of HSI and LiDAR (HSI-LiDAR) data, respectively. However, this kind of single-network architecture has limited ability to represent complex features, which may deliver a suboptimal classification result. Moreover, many existing hybrid network frameworks often rely on simple feature fusion operations, such as concatenation, cascading, or summation, and fail to explicitly model the complex complementary relationships across modalities. To address this issue, this article proposes a spatial–spectral multiscale Mamba (S2MSMamba) fusion network for HSI-LiDAR data classification. In the proposed framework, the Mamba branch models the global long-range dependencies with linear computational complexity, while the convolutional neural network (CNN) branch extracts local spatial textures and edge information. S2MSMamba jointly exploits the complementary strengths of Mamba and CNN by effectively mining the global long-range dependencies and the local spatial structure features of ground objects. In addition, a multiscale hierarchical alignment mechanism is developed to address scale variations of ground objects. Moreover, an adaptive decision-level fusion strategy is further adopted to integrate unimodal feature to refine classification performance. Experiments conducted on the Houston, MUUFL, and Trento benchmark datasets show that the proposed S2MSMamba outperforms existing methods on several evaluation metrics, demonstrating its effectiveness for land-cover classification The proposed method and the data used in this article are available at https://github.com/binzhao1018/S2MSMamba.**

***Index Terms*—Decision-level fusion, hyperspectral image (HSI) classification, light detection and ranging (LiDAR), Mamba, multimodal remote sensing data fusion.**

Received 29 June 2026; revised 26 July 2026 and 21 August 2026; accepted 2 September 2026. Date of publication 7 September 2026; date of current version 28 September 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 42401467 and Grant 62501327; in part by the Natural Science Foundation of Shandong Province, China, under Grant ZR2025MS521; in part by the Open Fund of State Key Laboratory of Remote Sensing Science under Grant OFSLRSS202415; and in part by Shandong Provincial Higher Education Youth Innovation Science and Technology Support Program, China, under Grant 2024KJH092. *(Corresponding author: Weiwei Song.)*

Bin Zhao is with the College of Information Science and Engineering, Shandong Agricultural University, Taian 271018, China, also with the State Key Laboratory of Remote Sensing Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100101, China, and also with the Faculty of Electrical and Computer Engineering, University of Iceland, 101 Reykjavik, Iceland (e-mail: binzhao@sdau.edu.cn).

Wenjie Cao is with the College of Information Science and Engineering, Shandong Agricultural University, Taian 271018, China (e-mail: 19819061372@163.com).

Lin Han is with the No.5 Geological Brigade of Shandong Provincial Bureau of Geology and Mineral Resources (No.5 Institute of Geology and Mineral Exploration of Shandong Province), Taian 271000, China and also with Shandong Provincial Laboratory of Big Data Development and Innovation for Geological Resources and Environment, Taian, Shandong 271000, China (e-mail: hanlin.geo363@gmail.com).

Weiwei Song is with the Pengcheng Laboratory, Shenzhen 518055, China (e-mail: weiweisong415@gmail.com).

Wenjuan Zhang is with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: zhangwj@aircas.ac.cn).

Lianru Gao is with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: gaolr@aircas.ac.cn).

Jakob Sigurdsson, Magnus O. Ulfarsson, and Jón Atli Benediktsson are with the Faculty of Electrical and Computer Engineering, University of Iceland, 101 Reykjavik, Iceland (e-mail: jakobs@hi.is; mou@hi.is; benedikt@hi.is).

Digital Object Identifier 10.1109/TGRS.2026.3731189

I. Introduction

SINCE the emergence of remote sensing technology in the 1960s, it has rapidly evolved into an interdisciplinary field and has played a central role in Earth observation. Owing to its large-scale, periodic, and noncontact observation capabilities, remote sensing enables efficient surface information acquisition and provides strong support for a wide range of Earth observation tasks, ranging from fundamental environmental monitoring and disaster assessment to sophisticated applications, such as land-cover classification in precision agriculture and spatial urban interpretation [1], [2], [3], [4], [5].

The richness of multiband remote sensing data significantly enhances land-cover separability and classification accuracy. Among them, hyperspectral image (HSI), as a critical data modality, has become an important branch of remote sensing research [6], [7], [8], [9], [10]. Hyperspectral sensors can acquire hundreds of contiguous narrow spectral bands, and their high spectral continuity and resolution enable fine characterization of subtle spectral variations between different types of land cover, providing a solid foundation for HSI classification [11], [12], [13], [14]. However, as a passive remote sensing technique, HSI is highly susceptible to atmospheric conditions and illumination variations, which often result in spectral instability [15], [16], [17]. Furthermore, in scenarios characterized by complex topography or highly heterogeneous spatial structures, spectral signatures alone are typically inadequate to fully capture the 3-D attributes of ground targets, thereby constraining further enhancements in classification accuracy [18].

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:08:01 UTC from IEEE Xplore. Restrictions apply.

In contrast, light detection and ranging (LiDAR), as an active remote sensing technology, can directly acquire high-precision elevation and spatial structural information, exhibiting distinct advantages in 3-D shape representation [19], [20]. Integrating LiDAR and HSI through multimodal data fusion can enhance land-cover discriminability while compensating for the limitations of HSI in spatial structure representation, thereby extending surface understanding from 2-D planar perception to 3-D spatial cognition. Although multimodal remote sensing data provide complementary information and improved classification performance, they also pose greater challenges to feature representation, information fusion strategies, and classification model design, which have become central research issues in HSI and LiDAR (HSI-LiDAR) data classification [21], [22].

Traditional machine learning methods for classification of fused HSI-LiDAR data have evolved from simple feature concatenation to more advanced feature modeling. Early study [23] demonstrated that LiDAR elevation information improves the discrimination of spectrally similar classes in classifiers such as support vector machines (SVMs) through direct feature concatenation. This improvement results from LiDAR providing elevation information, which explicitly distinguishes objects with identical spectral signatures but different physical elevations. These methods based on simple feature concatenation usually result in information redundancy. To overcome this problem, subsequent methods have been proposed by incorporating spatial prior information. Khodadadzadeh et al. [24] employed attribute profiles (APs) [25] and extended multi-APs (EMAPs) [26] to model spatial features, and proposed a probabilistic fusion framework combining subspace multinomial logistic regression (MLRsub) with Markov random fields (MRFs). Xia et al. [27] integrated morphological features with dimensionality reduction techniques such as principal component analysis (PCA) and adopted a random forest-based ensemble strategy to improve classification performance for complex land-cover types research area. Despite their effectiveness, the aforementioned methods generally rely heavily on high-quality and high-density HSI–LiDAR data, leading to high data acquisition costs. Moreover, their generalization capability remains limited in scenarios involving classes with few training samples and severe interclass similarity. Overall, traditional machine learning methods depend excessively on handcrafted feature design and lack the ability to automatically learn high-level discriminative representations from data, resulting in limited model expressiveness and generalization performance [28], [29]. These limitations have motivated a gradual transition from traditional machine learning methods toward deep learning-based methods.

Deep learning-based HSI-LiDAR data classification methods can generally be divided into convolutional neural network (CNN)-based methods [30], [31], Transformer-based methods [32], [33], and hybrid network-based methods [34]. The network architectures of CNN-based multimodal fusion classification methods are progressively evolving from shallow feature coupling to deep discriminative modeling. Hang et al. [35] proposed a foundational framework referred to as CoupledCNN for dual-branch fusion through high-level parameter sharing, yet it remains heavily dependent on labeled data. To alleviate the aforementioned label scarcity, Zhang et al. [36] proposed the IP-CNN for multimodal fusion. IP-CNN is an unsupervised method that can effectively reduce the annotation demands. However, the aforementioned advantage also limits the feature discriminability of IP-CNN. Song et al. [37] constructed a hashing-based deep metric learning (HDML) framework to explicitly constrain feature distribution, which significantly enhances feature discriminability.

With the advancement of self-attention mechanisms, Transformer-based multimodal remote sensing classification methods have gradually emerged and exhibited a clear progressive evolutionary trend. Currently, Transformer-based architectures for multimodal classification are evolving toward deep fusion and computational efficiency. Xue et al. [38] proposed a basic hierarchical cross-attention fusion framework, which establishes a foundational framework for global fusion of multimodal heterogeneous data. Furthermore, Zhang et al. [39] introduced a local-based multisource feature interactor (L-MSFI), which mitigates feature homogenization and information redundancy through local fusion and dynamic selection. To overcome the high computational bottleneck of self-attention, Zhao et al. [40] utilized the fractional Fourier image transformer (FrIT) to replace self-attention to efficiently capture the global contexts and enable multimodal fusion.

To overcome the inherent limitations of CNNs in global modeling and Transformers in local perception, recent studies have increasingly adopted hybrid network architectures for multimodal data fusion. Hybrid networks for multimodal fusion are evolving from basic decision-level integration to deep multimodal feature fusion. For instance, Ding et al. [41] performed decision-level fusion of local spatial and global spectral features using CNN and Transformer (GLT-Net). Moving toward deeper feature-level integration, Roy et al. [42] encoded LiDAR data as external tokens to guide multimodal fusion through a cross-attention mechanism (MFT). To further overcome the limitations of regular grids, Wang and Wang [43] developed a dual-coupled CNN-GCN network. This architecture combines the short-range perception of CNN with the medium-to-long-range topology-capturing capabilities of GCN, which demonstrates the significant potential of hybrid networks in processing complex land-cover spatial structures and facilitating deep multimodal fusion.

Despite the remarkable success of the aforementioned deep learning architectures in multimodal remote sensing classification, they still face inherent bottlenecks. Specifically, CNN-based methods inherently focus on local receptive field modeling and are limited in capturing the global spatial–spectral dependencies. Although Transformers possess powerful long-range modeling capabilities, their computational complexity grows quadratically with sequence length, resulting in large parameter sizes and high training and inference costs. While hybrid networks attempt to balance these two paradigms, they often suffer from increased model complexity and cumbersome multimodal fusion mechanisms. Consequently, their application in high-resolution and large-scale remote sensing scenarios is still subject to certain limitations.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:08:01 UTC from IEEE Xplore. Restrictions apply.

In this context, the Mamba architecture based on a state-space model (SSM) [44] effectively captures long-range dependencies with linear computational complexity, which provides an efficient paradigm for remote sensing classification [45]. By reformulating an HSI into long-sequence representations, Mamba can effectively characterize high-dimensional spectral features and their long-range dependencies, while avoiding the computational and memory bottlenecks associated with CNNs and Transformers. Consequently, Mamba provides an efficient and scalable solution for large-scale HSI-LiDAR data modeling in high-resolution and multimodal scenarios. Recently, researchers have begun to explore the potential of Mamba in remote sensing. Huang et al. [46] proposed a spectral–spatial dual-branch Mamba network for HSI classification. Furthermore, in the context of multimodal remote sensing, Liao et al. [47] proposed HLMamba, which ventures into the joint classification of HSI-LiDAR data by modeling multimodal long-range dependencies. However, the inherent 1-D scanning mechanism of Mamba limits its ability to capture critical local structures and edge details in LiDAR data. Thus, it is essential to address the lack of local inductive bias in order to advance multimodal data fusion.

In this article a spatial–spectral multiscale Mamba (S2MSMamba) fusion network for HSI-LiDAR data classification is proposed. S2MSMamba is a novel dual-branch architecture based on Mamba and CNN. Specifically, the Mamba architecture leverages its global context modeling capability to capture long-range dependencies, which excavates global correlations within high-dimensional spectral sequences. The CNN utilizes its convolutional architecture to extract local spatial structures from HSI-LiDAR data and detects high-frequency edge features from LiDAR data. Thus, S2MSMamba integrates the benefits of Mamba and CNN to extract the global–local, spatial–spectral, and high-frequency textural features from HSI-LiDAR data simultaneously. Furthermore, a multiscale hierarchical alignment mechanism is introduced to effectively accommodate the scale variations of diverse ground targets by facilitating cross-scale feature aggregation. Meanwhile, an adaptive decision-level fusion strategy is further employed to dynamically balance multimodal and unimodal features, which mitigates the over-suppression of weaker modalities and maximizes overall classification reliability. Therefore, S2MSMamba can yield effective classification performance. The main contributions of this article can be summarized as follows.

1) An S2MSMamba is developed for HSI-LiDAR data classification. S2MSMamba establishes a global–local representation architecture for heterogeneous modalities, where Mamba captures long-range spatial–spectral dependencies and a parallel CNN residual branch preserves local spatial structures and boundary details.
2) S2MSMamba develops modality-specific feature extraction mechanisms and a comprehensive multimodal fusion strategy to better capture the heterogeneous properties of different modalities. Specifically, a joint spatial–spectral Mamba block and a customized spatial Mamba block are constructed for HSI-LiDAR, respectively. Furthermore, multiscale hierarchical alignment and adaptive decision-level fusion are utilized to improve the classification performance.
3) To validate the performance of S2MSMamba, extensive experiments are conducted on three HSI-LiDAR datasets: Houston, MUUFL, and Trento. The results demonstrate that S2MSMamba outperforms other state-of-the-art methods in terms of overall accuracy (OA), average accuracy (AA), and the Kappa coefficient.

This article is structured as follows. Section II reviews related work, and Section III introduces preliminary knowledge. Section IV details the proposed framework. Section V presents the experimental results and analysis, and in Section VI, conclusions are drawn.

II. Related Work

In this article, we propose the S2MSMamba method, which is a dual-branch network architecture dedicated to the fusion and classification of HSI-LiDAR data. By synergistically integrating the Mamba and CNN framework, this method effectively leverages the complementary strengths of diverse network paradigms. To contextualize the proposed method, Sections II-A and II-B review the existing literature from two primary perspectives: the application of dual-branch networks in multimodal data fusion, and the recent evolution of Mamba architectures.

A. Dual-Branch Networks in Multimodal Data Fusion

In the field of multimodal remote sensing data fusion, dual-branch networks have evolved from feature concatenation to deep fusion and global modeling. Early methods mainly relied on feature concatenation. For instance, Xu et al. [48] utilized 1-D and 2-D CNNs to independently extract HSI-LiDAR features and fused them at high semantic levels. However, such methods show limited capacity in modeling multimodal associations. To enhance multimodal fusion, subsequent studies introduced more tightly coupled and discriminative feature learning mechanisms. Hang et al. [35] proposed CoupledCNN, which achieves multimodal guidance through parameter sharing. Wu et al. [49] designed a cross-channel reconstruction (CCR) network, which improves the compactness of multimodal feature representations. Song et al. [50] introduced a deep hashing mechanism, which constrains multimodal feature similarity. Zhao et al. [51] utilized a hierarchical random walk strategy to enhance spatial consistency. These methods improve multimodal correlation modeling, but challenges remain in adaptive representation of heterogeneous features and long-range dependency modeling. With the development of Transformer architectures, fusion methods have been proposed by incorporating the global attention mechanisms. Yang et al. [52] proposed a cross-attention-based fusion Transformer, which enables global feature fusion but has limited capability in modeling local spatial structures. In addition, to balance local and global information, hybrid architectures have been developed. Zhao et al. [53] combined CNN and Transformer to achieve multimodal fusion via cross-token attention, but the computational cost remains high. Sun et al. [54] further introduced a lightweight Transformer structure, which reduces

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:08:01 UTC from IEEE Xplore. Restrictions apply.

computational complexity, but long-range dependency modeling is not fully addressed. In summary, existing dual-branch fusion methods have evolved from simple feature concatenation to deep fusion and global modeling in multimodal feature representation and have improved the utilization of multimodal information to some extent. However, there is still an open problem of how to achieve adaptive representation of complex heterogeneous features and effective modeling of long-range dependencies under computational efficiency constraints.

B. Mamba Networks in Remote Sensing

In recent years, the Mamba architecture has rapidly evolved in remote sensing, with the primary advances along two trajectories: single-modal HSI interpretation and multimodal data fusion. To enhance single-modal HSI classification, researchers have focused on improving the internal scanning mechanisms of Mamba and extending its dimensionality. For instance, He et al. [55] proposed 3DSS-Mamba, which extends the SSM to 3-D hyperspectral cubes via a 3-D selective scanning mechanism to achieve efficient global representation. To strengthen spatial structure perception, Wang et al. [56] developed LE-Mamba, which integrates multidirectional local scanning to jointly model 2-D spatial structures and 1-D spectral information. Similarly, Wang et al. [57] introduced S2Mamba, which decouples spatial and spectral scanning and utilizes a gating mechanism to adaptively fuse dual-path features. Moreover, Ahmad et al. [58] proposed WaveMamba, which integrates wavelet transforms to process hyperspectral tokens for multiresolution modeling of local textures and global contexts. Alongside internal mechanism improvements, hybrid architectures have emerged to address the limitations of Mamba in capturing local details. Sheng et al. [59] designed DualMamba, which is a parallel dual-stream network that combines Mamba with a lightweight CNN to capture both global dependencies and local spatial details in single-modal HSI. To address spectral redundancy and complex spatial modeling challenges, Yang et al. [60] proposed a novel paradigm that couples graph structure learning with Vision Mamba. Furthermore, researchers have explicitly designed multimodal interactive mechanisms to extend Mamba into multimodal data fusion. Gao et al. [61] introduced MSFMamba to utilize a dual-input fusion module to achieve selective multimodal feature integration. Despite these remarkable advancements, current multimodal Mamba architectures generally rely on simple feature concatenation strategies. Crucially, they fail to explicitly compensate for the inherent deficiency of Mamba in local spatial information bias, which remains essential for effectively modeling the spatial geometric structures from LiDAR data. Therefore, it is an urgent and unresolved challenge to effectively couple the global long-range modeling capabilities of Mamba with the local structural perception of CNN within a unified multimodal fusion framework.

III. Preliminary Knowledge

This section provides a brief overview of the theoretical foundations underlying the proposed framework, which focuses specifically on the SSM and the Mamba architecture.

Mamba is an efficient architecture based on the SSM. The theoretical foundation of Mamba is derived from the classical SSM, which was originally proposed to characterize continuous-time, linear, and time-invariant dynamical systems. The mathematical representation of the model is expressed as

$$\begin{cases} h'(t) = Ah(t) + Bx(t) \\ y(t) = Ch(t) \end{cases} \tag{1}$$

where $h(t) \in \mathbb{R}^N$ is the hidden state, $x(t) \in \mathbb{R}$ is the input, $y(t) \in \mathbb{R}$ is the output, and $A \in \mathbb{R}^{N\times N}$, $B \in \mathbb{R}^{N\times 1}$, and $C \in \mathbb{R}^{1\times N}$ are the system parameter matrices.

To apply continuous-time SSM to discrete sequence modeling, the zero-order hold (ZOH) method is employed for discretization, which results in the structured state-space sequence (S4) model. Under a fixed time step Δ, the discretized system parameters $\bar{A}, \bar{B}$ can be approximately expressed as

$$\begin{cases} \bar{A} = \exp(\Delta A) \\ \bar{B} = (\Delta A)^{-1}\left(\exp(\Delta A) - I\right) \cdot \Delta B \\ \quad \approx (\Delta A)^{-1}(\Delta A)(\Delta B) \\ \quad = \Delta B. \end{cases} \tag{2}$$

Thus, the discrete-time state update equation can be denoted as

$$\begin{cases} h_k = \bar{A}h_{k-1} + \bar{B}x_k \\ y_k = Ch_k. \end{cases} \tag{3}$$

As the S4 parameters $(\bar{A}, \bar{B}, C)$ are input-independent, the overall computation can be equivalently rewritten as a global convolution operation

$$y = x * \bar{K} \tag{4}$$

where $*$ denotes the discrete linear convolution operator, and $\bar{K} = (C\bar{B}, C\bar{A}\bar{B}, \ldots, C\bar{A}^{R-1}\bar{B})$ is the structured convolutional kernel of length R (i.e., the length of the input sequence x). This equivalence reveals the intrinsic connection between SSM and CNN, i.e., an SSM can be regarded as an infinitely long, parameter-shared, and structured convolutional kernel capable of capturing global dependencies, whereas conventional CNNs are constrained by local receptive fields and fixed kernel sizes. However, the parameter matrices A, B, and C in S4 remain static throughout the entire sequence processing, which lacks sensitivity to input content and thus limits their ability to model context-dependent dynamic dependencies.

To address this limitation, the selective SSM model further introduces an input-dependent selective mechanism, which reformulates the system parameters as learnable functions of the input. Specifically, $A(x)$, $B(x)$, and $C(x)$ are adaptively modulated according to the current input sequence. This design enables the model to dynamically attend to salient information while suppressing irrelevant interference based on contextual cues, which substantially enhances its capacity to model local structures and semantic features.

Mamba, based on the selective SSM framework, constitutes an efficient paradigm for sequence modeling. By integrating selective state-space mechanisms with hardware-aware parallel scan algorithms, Mamba preserves strong long-range

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:08:01 UTC from IEEE Xplore. Restrictions apply.

dependency modeling capability while achieving high-fidelity perception of local patterns to attain an excellent balance between computational efficiency and modeling performance.

The relationship between the input x and output x_{out} of the Mamba can be expressed as

$$x_{out} = \text{Mamba}(x) \tag{5}$$

where Mamba$(\cdot)$ is the Mamba function. To better understand the internal computation process, it can be expressed as

$$\begin{cases} z_1 = \text{SSM}\left(\sigma\left(f_{\text{conv}}\left(f_{\text{linear}}(x)\right)\right)\right) \\ z_2 = \sigma\left(f_{\text{linear}}(x)\right) \\ x_{out} = f_{\text{linear}}\left(z_1 \odot z_2\right). \end{cases} \tag{6}$$

where $\sigma(\cdot)$ denotes the sigmoid linear unit (SiLU) activation function. $f_{\text{linear}}(\cdot)$ and $f_{\text{conv}}(\cdot)$ represent the linear projection and causal convolution operations, respectively. SSM$(\cdot)$ denotes the selective SSM, and the operator $\odot$ denotes element-wise multiplication.

IV. Proposed Framework

To address the characteristics of HSI-LiDAR data in multimodal remote sensing classification, this article proposes S2MSMamba, based on a dual-branch architecture of Mamba and CNN. This framework aims to capture global sequence dependencies with linear complexity using Mamba, which simultaneously extracts local spatial features via a parallel CNN residual branch. The overall network of S2MSMamba comprises three core stages, i.e., dual-branch feature extraction, multiscale multimodal feature fusion, and adaptive decision-level fusion classification. The framework of S2MSMamba is shown in Fig. 1.

A. Dual-Branch Feature Extraction

To capture the distinct physical properties of multimodal data, we design a unified dual-branch feature extraction framework, which effectively couples the global long-range modeling of Mamba with the local structural perception of CNN.

Let $\mathcal{X}_i \in \mathbb{R}^{C_i \times H \times W}$ denote the input image patch, where $i \in \{h, l\}$, when $i = h$, $\mathcal{X}_i$ is the patch of HSI, and $i = l$, $\mathcal{X}_i$ is the patch of LiDAR. To fully exploit complementary characteristics, the feature extraction for each modality is symmetrically parallelized into a global Mamba branch and a local CNN residual branch.

1) Global Dependency Modeling via Mamba Branch: The input $\mathcal{X}_i$ is first projected into a D-dimensional hidden space to obtain a shallow feature tensor $\mathcal{F}_i^{(0)} \in \mathbb{R}^{D \times H \times W}$

$$\mathcal{F}_i^{(0)} = \sigma\left(\text{GN}\left(\text{Conv}_{1\times1}\left(\mathcal{X}_i\right)\right)\right) \tag{7}$$

where GN represents group normalization.

For spatial modeling across both modalities, the tokenization $\mathcal{T}_{spa}$ flattens $\mathcal{F}_i^{(0)}$ into a 1-D sequence. Taking this flattened sequence as input, the spatial representation $\mathcal{F}_i^{spa}$ is extracted via the spatial Mamba block

$$\mathcal{F}_i^{spa} = \mathcal{F}_i^{(0)} + \text{Proj}\left(\mathcal{T}_{spa}^{-1}\left(\text{Mamba}\left(\mathcal{T}_{spa}\left(\mathcal{F}_i^{(0)}\right)\right)\right)\right) \tag{8}$$

where Mamba$(\cdot)$ denotes the forward operation of the Mamba block and $\mathcal{T}_{spa}^{-1}$ restores the 2-D spatial structure, and Proj$(\cdot)$ denotes a feature projection operation consisting of GN and a SiLU activation function.

Similarly, for HSI spectral modeling, the tokenization $\mathcal{T}_{spe}$ expands the channel dimension via zero-padding to D_{pad} and groups it into N segments, which yields a sequence representation. Using these grouped segments as an input sequence, the spectral representation $\mathcal{F}_h^{spe}$ is computed via the spectral Mamba block

$$\mathcal{F}_h^{spe} = \mathcal{F}_h^{(0)} + \text{Proj}\left(\mathcal{T}_{spe}^{-1}\left(\text{Mamba}\left(\mathcal{T}_{spe}\left(\mathcal{F}_h^{(0)}\right)\right)\right)\right) \tag{9}$$

where $\mathcal{T}_{spe}^{-1}$ restores the spectral structure.

For the HSI branch, the spatial and spectral representations are adaptively fused via learnable Softmax weights (w_1, w_2), whereas the LiDAR branch preserves only the spatial features. Thus, the fused features $\mathcal{F}_i^{fuse}$ are

$$\mathcal{F}_i^{fuse} = \begin{cases} w_1\mathcal{F}_h^{spa} + w_2\mathcal{F}_h^{spe}, & i = h \\ \mathcal{F}_l^{spa}, & i = l. \end{cases} \tag{10}$$

Defining a standard convolutional operator $\mathcal{C}(\mathbf{x}) = \delta(\text{BN}(\text{Conv}_{3\times3}(\mathbf{x})))$ with a ReLU activation function $\delta(\cdot)$, the refined Mamba feature $\mathcal{F}_i^M$ is obtained through a sequential cascade

$$\mathcal{F}_i^M = \text{Drop}\left(\mathcal{C}\left(\mathcal{C}\left(\mathcal{F}_i^{fuse}\right)\right)\right) \tag{11}$$

where Drop$(\cdot)$ denotes the Dropout operation for improving generalization and mitigating overfitting.

2) Local Perception via Parallel CNN Residual Branch: To preserve high-frequency edge details and spatial textures, a parallel CNN branch processes the original input $\mathcal{X}_i$, initialized by $\mathcal{H}_i^{(0)} = \mathcal{C}(\mathcal{X}_i)$.

We deploy a cascade of $Q = 3$ CNN residual blocks to mine local spatial structures. Residual block q is explicitly defined as

$$\mathcal{H}_i^{(q)} = \delta\left(\mathcal{H}_i^{(q-1)} + \text{Drop}\left(\widetilde{\mathcal{C}}\left(\mathcal{C}\left(\mathcal{H}_i^{(q-1)}\right)\right)\right)\right) \tag{12}$$

where $\widetilde{\mathcal{C}}(\mathbf{x}) = \text{BN}(\text{Conv}_{3\times3}(\mathbf{x}))$. Following the cascade, the output is aligned to match the Mamba path dimensions, which yields the local CNN feature $\mathcal{F}_i^C$:

$$\mathcal{F}_i^C = \text{Drop}\left(\mathcal{C}\left(\mathcal{H}_i^{(Q)}\right)\right). \tag{13}$$

3) Intramodal Feature Fusion: The global long-range semantic dependencies and local spatial structural features are seamlessly integrated via element-wise addition. The final modality-specific representation $\mathcal{F}_i$ is symmetrically constructed as

$$\mathcal{F}_i = \mathcal{F}_i^M + \mathcal{F}_i^C, \quad i \in \{h, l\}. \tag{14}$$

B. Multiscale Multimodal Feature Fusion

To address the scale variations of ground targets, the feature extraction framework detailed above is applied in parallel to input patches at two distinct spatial scales: $p \times p$ and $2p \times 2p$.

Let $\mathcal{F}_{i,p}$ and $\mathcal{F}_{i,2p}$ $(i \in \{h, l\})$ denote the extracted modality-specific features at their respective scales. To enable cross-scale fusion, the larger-scale features $2p \times 2p$ are spatially aligned to the $p \times p$ resolution using adaptive average

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:08:01 UTC from IEEE Xplore. Restrictions apply.

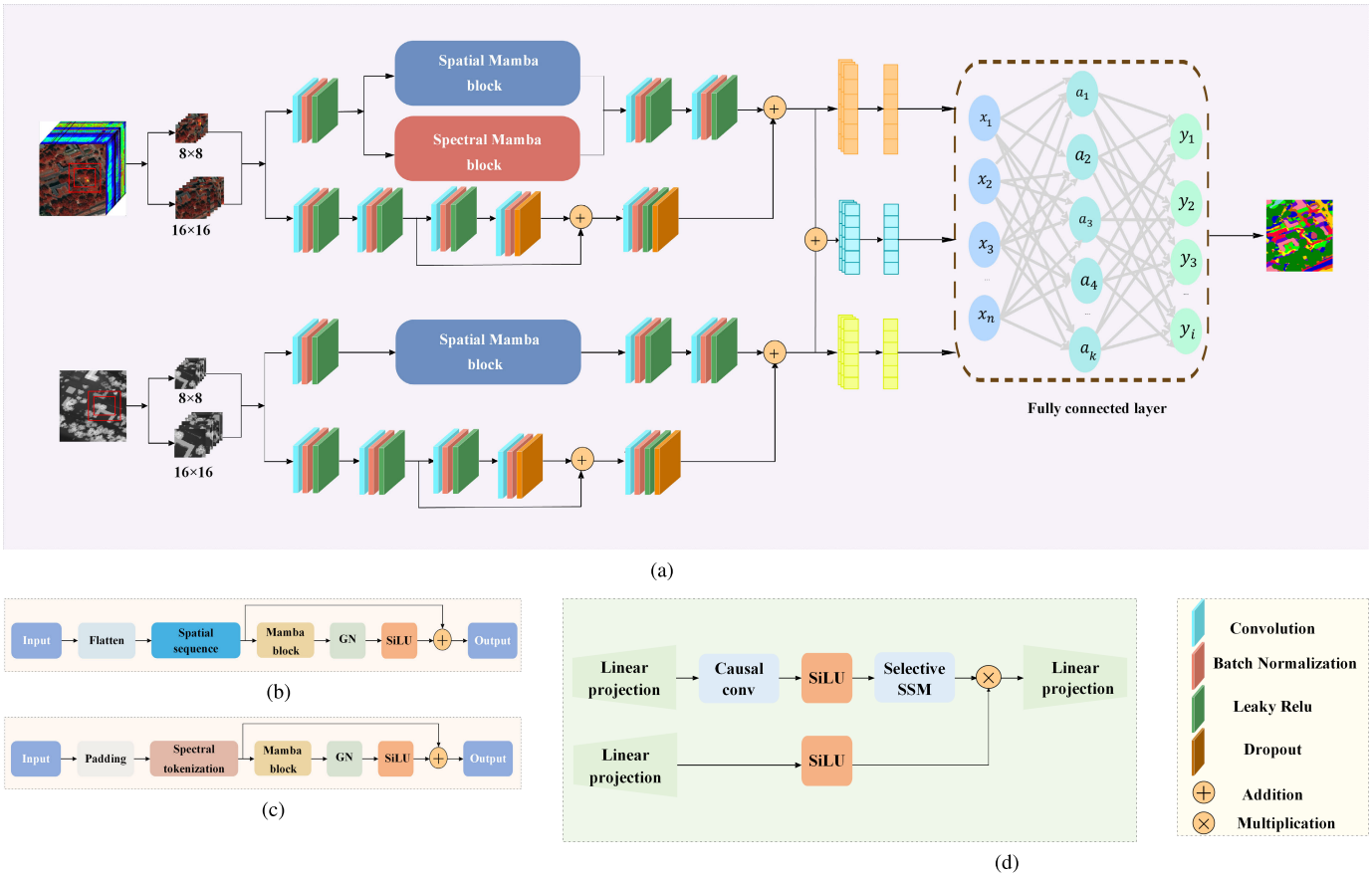


Fig. 1. Overall structure of our proposed S2MSMamba framework. (a) Overall framework with hyperspectral and LiDAR images as inputs and a classification map as the output, (b) spatial Mamba block, (c) spectral Mamba block, and (d) Mamba block.

pooling (AAP), denoted by a prime symbol. Therefore, $\mathcal{F}'_{i,2p}$ is symmetrically constructed as

$$\mathcal{F}'_{i,2p} = \mathrm{AAP}\left(\mathcal{F}_{i,2p}\right), \quad i \in \{h, l\}. \tag{15}$$

At each scale, the HSI-LiDAR features are fused through element-wise multiplication ($\odot$) to facilitate intermodal fusion. The scale-specific fused features are then aggregated via element-wise addition to yield the comprehensive cross-scale fused tensor $\mathcal{X}_f$

$$\mathcal{X}_f = \left(\mathcal{F}_{h,p} \odot \mathcal{F}_{l,p}\right) + \left(\mathcal{F}'_{h,2p} \odot \mathcal{F}'_{l,2p}\right). \tag{16}$$

To transition from spatial feature maps to semantic vectors for downstream classification, global average pooling (GAP) compresses the 3-D feature tensor into a 1-D global feature $\mathbf{g}_m$ ($m \in \{f, h, l\}$). Specifically, the fused feature $\mathbf{g}_f$ and the robust unimodal feature $\mathbf{g}_m$ are, respectively, computed as

$$\mathbf{g}_f = \mathrm{GAP}\left(\mathcal{X}_f\right) \tag{17}$$

$$\mathbf{g}_i = \frac{1}{2}\left(\mathrm{GAP}\left(\mathcal{F}_{i,p}\right) + \mathrm{GAP}\left(\mathcal{F}'_{i,2p}\right)\right), \quad i \in \{h, l\}. \tag{18}$$

C. Adaptive Decision-Level Fusion Classification

To maximize classification reliability and prevent the dominant modality from over-suppressing the weaker one, an adaptive decision-level fusion mechanism is formulated based on the extracted global features.

First, to generate the classification probabilistic prediction $\mathbf{y}_m$, the global feature $\mathbf{g}_m$ is projected to a unified dimension via a modality-specific weight matrix $\mathbf{W}_m$, followed by Drop, $m \in \{f, h, l\}$ corresponds to the fused feature, HSI feature, and LiDAR feature, respectively. Subsequently, a shared classification head maps $\mathbf{g}_m$ to the class space using layer normalization (LN) and a linear classifier. This sequential operation is efficiently formulated as

$$\mathbf{y}_m = \mathbf{W}_{cls}\mathrm{LN}\left(\mathrm{Drop}\left(\mathbf{W}_m\mathbf{g}_m\right)\right) + \mathbf{b}_{cls}, \quad m \in \{f, h, l\} \tag{19}$$

where $\mathbf{W}_{cls}$ and $\mathbf{b}_{cls}$ are, respectively, the weights and biases of the linear classifier, which are strictly shared across all three branches to prevent overfitting and enhance generalization.

The final classification prediction $\mathbf{y}_{out}$ is determined by a learnable weighted sum of these three independent classification features

$$\left[\omega_f, \omega_h, \omega_l\right] = \mathrm{Softmax}\left(\left[\alpha_f, \alpha_h, \alpha_l\right]\right) \tag{20}$$

$$\mathbf{y}_{out} = \omega_f\mathbf{y}_f + \omega_h\mathbf{y}_h + \omega_l\mathbf{y}_l \tag{21}$$

where $\boldsymbol{\alpha} = [\alpha_f, \alpha_h, \alpha_l] \in \mathbb{R}^3$ are learnable scalar parameters. These parameters adaptively learn the fusion weights of the predictions from different branches, thereby balancing multimodal evidence and complementary unimodal information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:08:01 UTC from IEEE Xplore. Restrictions apply.

V. Experimental Results and Analysis

This section provides a comprehensive evaluation of the performance of the proposed S2MSMamba method. First, the experimental process begins with a detailed description of the experimental settings, the HSI-LiDAR datasets, followed by comparison methods. Then, the proposed method is thoroughly compared with state-of-the-art methods from both quantitative and qualitative perspectives, which include classification accuracy comparisons and visual analysis of classification maps. Finally, ablation experiments are conducted to further investigate the contribution of each key module to the overall classification performance, which thereby validates the effectiveness and robustness of the proposed S2MSMamba method for multimodal HSI-LiDAR data classification task.

A. Experimental Settings

To verify the effectiveness of the proposed S2MSMamba method, extensive experiments are conducted on three publicly available HSI-LiDAR datasets. All experiments are carried out on the same server platform running Ubuntu 22.04, and S2MSMamba is implemented using the PyTorch deep learning framework. The hardware environment consists of an NVIDIA GeForce RTX 4090 GPU and an Intel[1] Xeon[1] Platinum 8368Q CPU with a clock speed of 2.60 GHz. In all experiments, the network is trained using the Adam optimizer with an initial learning rate of 5×10^{-4}. The maximum number of training epochs is set to 500, and the batch size is fixed at 64. To effectively exploit the spatial contextual information of HSI, spatial patches with sizes of 8×8 and 16×16 (i.e., setting the scale parameter $p = 8$) are used as the network inputs. For data partitioning, a stratified random sampling strategy is adopted, where 20 samples per class are randomly selected as the training set for each dataset, while the remaining samples are used for testing. The detailed sample distribution of experimential hy perspectral datasets is shown in Table I. To eliminate the influence of random network initialization and sample selection, all experiments are independently repeated 10 times with different random seeds, and the mean values along with their standard deviations are reported. To comprehensively evaluate the classification performance, OA, AA, and the Kappa coefficient (Kappa) are adopted as evaluation metrics. Specifically, OA measures the overall classification accuracy, AA reflects the average classification performance across different classes, and the Kappa evaluates the agreement between the predicted labels and the ground truth beyond chance. To further assess the generalization capability of the proposed method across different scenarios, three representative HSI-LiDAR datasets are selected and introduced in detail in the following subsection.

B. Datasets

1) Houston Dataset: The Houston dataset was acquired over the main campus of the University of Houston and its surrounding urban areas in Houston, Texas, USA. The study area is a complex urban scene with substantial scene diversity, which contains 15 land-cover classes. The size of the imagery is 349×1905 pixels with a spatial resolution of 2.5 m, and there are 144 hyperspectral bands along with a corresponding single-band LiDAR dataset. Fig. 2 illustrates (a) an RGB image created using a color matching function [62], (b) a grayscale image of LiDAR, and (c) the corresponding ground-truth map.

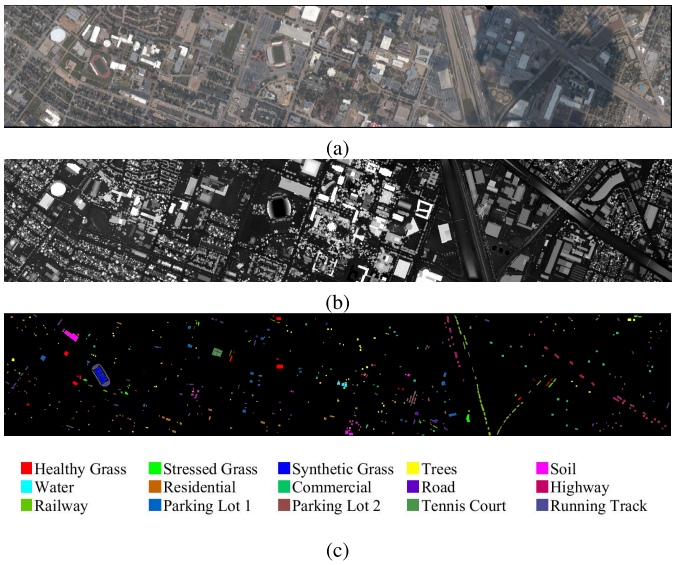


Fig. 2. Houston dataset. (a) RGB image, (b) grayscale image of LiDAR, and (c) ground-truth map.

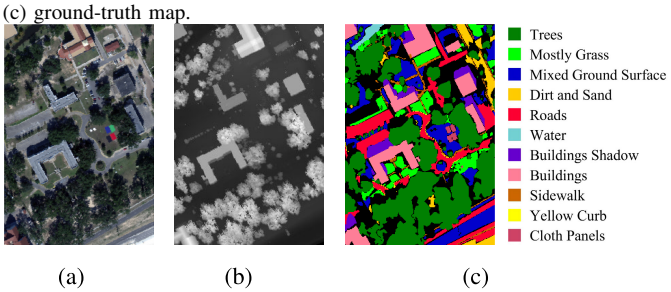


Fig. 3. MUUFL dataset. (a) RGB image, (b) grayscale image of LiDAR, and (c) ground-truth map.

2) MUUFL Dataset: The MUUFL dataset was acquired in the Gulfport area of Mississippi, USA. The study area exhibits rich spatial details and multimodal feature information, which includes a total of 11 land-cover classes. The size of the imagery is 325×220 pixels with a spatial resolution of 1 m, and there are 64 hyperspectral bands along with corresponding two-band LiDAR data. More details are shown in Fig. 3.

3) Trento Dataset: The Trento dataset was acquired in the suburban area of the city of Trento, Italy. The study area is a medium-scale scene mainly covering mixed urban and natural environments, which contains six land-cover classes with a relatively concentrated class distribution. The size of the imagery is 166×600 pixels with a spatial resolution of 1 m, and there are 63 hyperspectral bands along with a corresponding single-band LiDAR dataset. More details are shown in Fig. 4.

C. Comparison Methods

To validate the effectiveness of the proposed S2MSMamba method, it is compared with several land-cover classification methods based on multimodal data fusion, which includes coupled CNN (CoupledCNN) [35], deep encoder–decoder

[1]Registered trademark.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:08:01 UTC from IEEE Xplore. Restrictions apply.

TABLE I

SAMPLE DISTRIBUTION FOR HYPERSPECTRAL DATASETS

No.	Houston				MUUFL				Trento			
	Category	Training	Test	Total	Category	Training	Test	Total	Category	Training	Test	Total
C1	Healthy grass	20	1231	1251	Trees	20	23226	23246	Apple Trees	20	4014	4034
C2	Stressed grass	20	1234	1254	Mostly grass	20	4250	4270	Buildings	20	2883	2903
C3	Synthetic grass	20	677	697	Mixed Ground Surface	20	6862	6882	Ground	20	459	479
C4	Trees	20	1224	1244	Dirt and Sand	20	1806	1826	Woods	20	9103	9123
C5	Soil	20	1222	1242	Road	20	6667	6687	Vineyard	20	10481	10501
C6	Water	20	305	325	Water	20	446	466	Roads	20	3154	3174
C7	Residential	20	1248	1268	Buildings Shadow	20	2213	2233				
C8	Commercial	20	1224	1244	Buildings	20	6220	6240				
C9	Road	20	1232	1252	Sidewalk	20	1365	1385				
C10	Highway	20	1207	1227	Yellow Curb	20	163	183				
C11	Railway	20	1215	1235	Cloth Panels	20	249	269				
C12	Parking Lot 1	20	1213	1233								
C13	Parking Lot 2	20	449	469								
C14	Tennis Court	20	408	428								
C15	Running Track	20	640	660								
	Total	300	14729	15029	Total	220	53467	53687	Total	120	30094	30214

TABLE II

QUANTITATIVE RESULTS (ACC% $\pm$ STD%) ON THE HOUSTON DATASET (BEST RESULTS DISPLAYED IN BOLD AND SECOND-BEST RESULTS UNDERLINED)

No.	CoupledCNN	EndNet	CCR	GLT	HCT	HLMamba	MGMNet	MCFNet	S3F2Net	DAHGMN	S2MSMamba
C1	97.08 ± 0.39	92.17 ± 3.95	**97.29 ± 3.71**	91.21 ± 3.68	93.27 ± 3.69	93.02 ± 4.26	92.19 ± 2.84	93.39 ± 3.53	94.76 ± 4.03	93.32 ± 3.73	94.22 ± 4.03
C2	94.74 ± 3.44	93.35 ± 3.93	93.49 ± 3.95	96.39 ± 4.36	96.30 ± 4.03	**96.85 ± 4.08**	96.83 ± 1.12	95.39 ± 4.17	96.33 ± 3.74	96.42 ± 3.66	95.73 ± 3.63
C3	**99.63 ± 0.18**	99.41 ± 0.41	97.21 ± 1.91	98.63 ± 1.05	99.29 ± 0.64	98.76 ± 0.90	97.05 ± 1.10	99.23 ± 0.46	99.43 ± 0.52	98.98 ± 0.60	98.66 ± 1.03
C4	**99.88 ± 0.11**	95.25 ± 2.33	96.13 ± 1.10	98.60 ± 1.93	98.77 ± 1.06	98.54 ± 1.85	98.19 ± 0.86	98.51 ± 1.75	98.68 ± 1.68	98.36 ± 1.01	98.95 ± 0.62
C5	**100.00 ± 0.00**	98.08 ± 1.68	97.96 ± 1.46	99.89 ± 0.13	99.97 ± 0.06	99.97 ± 0.10	99.60 ± 0.25	99.98 ± 0.03	99.35 ± 1.08	**100.00 ± 0.00**	**100.00 ± 0.00**
C6	87.54 ± 3.32	91.38 ± 4.92	90.96 ± 4.58	96.07 ± 5.59	98.03 ± 4.12	95.85 ± 5.41	**98.79 ± 0.93**	96.26 ± 4.26	95.70 ± 4.30	97.90 ± 3.91	98.43 ± 3.91
C7	97.58 ± 1.17	83.71 ± 4.61	92.66 ± 1.76	94.73 ± 1.42	92.54 ± 3.58	96.84 ± 2.48	96.15 ± 0.62	96.97 ± 1.36	93.79 ± 3.50	96.75 ± 1.12	**99.13 ± 0.80**
C8	**94.18 ± 2.38**	85.35 ± 5.80	88.38 ± 6.57	88.19 ± 6.12	87.16 ± 4.97	88.90 ± 5.11	90.27 ± 2.15	90.10 ± 5.17	85.22 ± 5.40	91.98 ± 3.77	89.56 ± 5.54
C9	76.80 ± 4.64	75.84 ± 4.69	83.03 ± 4.26	89.41 ± 3.80	81.12 ± 6.63	91.41 ± 2.43	88.23 ± 2.90	91.70 ± 3.92	85.86 ± 4.02	90.58 ± 3.74	**92.25 ± 2.91**
C10	86.45 ± 0.74	75.98 ± 8.05	72.62 ± 2.43	93.24 ± 3.21	93.10 ± 6.08	95.61 ± 3.31	95.35 ± 5.55	93.99 ± 3.48	**97.19 ± 2.14**	93.08 ± 5.00	96.66 ± 2.24
C11	93.88 ± 1.72	87.42 ± 4.90	79.90 ± 3.89	96.30 ± 1.79	98.41 ± 1.21	96.31 ± 2.51	**98.98 ± 0.64**	96.68 ± 1.31	96.79 ± 1.21	97.84 ± 0.69	97.98 ± 1.27
C12	88.34 ± 2.65	71.66 ± 5.00	70.78 ± 4.91	89.16 ± 5.00	**92.36 ± 5.90**	89.97 ± 5.78	87.61 ± 2.69	90.68 ± 6.23	91.88 ± 5.10	88.96 ± 5.40	87.78 ± 4.52
C13	**99.87 ± 0.18**	58.06 ± 5.27	85.10 ± 5.27	98.57 ± 1.96	97.84 ± 2.01	97.87 ± 1.96	93.65 ± 2.03	96.93 ± 2.05	95.68 ± 2.59	97.84 ± 1.82	98.17 ± 2.54
C14	**100.00 ± 0.00**	97.77 ± 1.16	88.89 ± 5.90	99.98 ± 0.08	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	99.93 ± 0.23	**100.00 ± 0.00**	99.98 ± 0.07
C15	**100.00 ± 0.00**	99.11 ± 0.52	96.30 ± 2.03	99.91 ± 0.30	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	99.98 ± 0.05	99.34 ± 0.78	99.84 ± 0.30	99.97 ± 0.09
OA	93.82 ± 0.64	86.69 ± 1.11	87.63 ± 1.35	94.58 ± 0.58	94.28 ± 0.92	95.31 ± 0.46	94.94 ± 0.75	95.42 ± 0.38	94.71 ± 0.77	95.46 ± 0.63	**96.09 ± 0.88**
AA	94.40 ± 0.51	86.97 ± 0.94	88.71 ± 1.22	95.35 ± 0.70	95.21 ± 0.79	95.85 ± 0.54	95.53 ± 0.60	95.99 ± 0.42	95.33 ± 0.74	96.12 ± 0.62	**96.71 ± 0.69**
Kappa × 100	93.32 ± 0.70	85.61 ± 1.19	86.62 ± 1.46	94.14 ± 0.63	93.81 ± 1.00	94.93 ± 0.50	94.53 ± 0.81	95.05 ± 0.41	94.28 ± 0.83	95.09 ± 0.68	**95.74 ± 0.90**

TABLE III

QUANTITATIVE RESULTS (ACC% $\pm$ STD%) ON THE MUUFL DATASET (BEST RESULTS DISPLAYED IN BOLD AND SECOND-BEST RESULTS UNDERLINED)

No.	CoupledCNN	EndNet	CCR	GLT	HCT	HLMamba	MGMNet	MCFNet	S3F2Net	DAHGMN	S2MSMamba
C1	88.14 ± 1.02	79.79 ± 3.06	**97.92 ± 0.46**	89.75 ± 3.09	84.75 ± 2.25	83.45 ± 0.48	90.33 ± 1.85	88.40 ± 2.02	90.36 ± 2.69	90.25 ± 1.48	89.64 ± 0.16
C2	76.78 ± 3.06	75.27 ± 7.59	55.14 ± 5.09	77.41 ± 5.06	75.08 ± 7.26	**83.88 ± 0.77**	75.99 ± 3.15	76.51 ± 8.63	71.79 ± 7.79	73.24 ± 8.59	78.03 ± 0.42
C3	60.26 ± 3.88	61.18 ± 6.87	69.13 ± 4.81	64.82 ± 7.38	62.67 ± 4.81	60.08 ± 1.43	64.26 ± 3.69	66.57 ± 5.70	69.37 ± 5.40	63.47 ± 7.54	**70.73 ± 0.71**
C4	88.69 ± 5.15	78.76 ± 5.82	60.97 ± 4.94	88.97 ± 3.52	91.82 ± 2.72	**96.70 ± 1.11**	78.15 ± 5.60	92.10 ± 5.73	88.69 ± 4.31	89.22 ± 3.03	95.55 ± 0.30
C5	84.06 ± 1.92	86.50 ± 1.95	**92.09 ± 2.41**	77.35 ± 4.52	74.64 ± 6.25	91.88 ± 0.26	81.46 ± 3.84	80.94 ± 5.55	84.74 ± 4.11	78.28 ± 5.75	78.08 ± 0.20
C6	**100.00 ± 0.00**	95.56 ± 2.49	59.06 ± 8.40	99.64 ± 0.54	99.57 ± 0.51	99.91 ± 0.12	99.53 ± 0.79	99.22 ± 1.34	99.62 ± 0.50	99.87 ± 0.27	98.81 ± 0.10
C7	80.89 ± 3.91	86.60 ± 5.64	38.06 ± 3.54	90.76 ± 4.03	85.93 ± 5.42	**94.13 ± 0.20**	86.51 ± 9.19	92.96 ± 2.94	83.49 ± 5.74	90.55 ± 3.52	90.55 ± 0.29
C8	**95.93 ± 1.08**	87.35 ± 3.18	92.22 ± 4.20	91.92 ± 2.93	90.09 ± 4.00	89.09 ± 0.21	91.25 ± 3.85	93.92 ± 2.11	90.22 ± 3.24	91.91 ± 1.09	91.45 ± 0.21
C9	60.34 ± 3.81	59.55 ± 5.03	35.07 ± 6.90	54.51 ± 9.67	50.72 ± 6.24	50.95 ± 3.00	54.48 ± 4.26	60.97 ± 10.00	45.45 ± 11.82	52.29 ± 6.71	**75.36 ± 0.57**
C10	83.93 ± 2.19	82.76 ± 5.47	7.21 ± 1.37	74.91 ± 9.05	79.88 ± 7.86	67.67 ± 1.69	70.06 ± 5.96	71.29 ± 15.57	60.98 ± 21.68	63.74 ± 23.25	**85.28 ± 0.39**
C11	94.30 ± 2.21	92.65 ± 2.04	47.61 ± 7.15	97.19 ± 2.39	97.55 ± 2.67	**99.48 ± 0.38**	92.05 ± 2.31	96.95 ± 1.95	96.51 ± 1.81	95.30 ± 2.37	98.84 ± 0.28
OA	83.18 ± 0.35	78.69 ± 1.85	75.48 ± 1.64	83.46 ± 1.23	80.10 ± 1.10	82.42 ± 0.16	83.38 ± 0.91	84.06 ± 1.11	84.00 ± 0.87	83.19 ± 0.77	**85.04 ± 0.11**
AA	83.03 ± 0.68	80.54 ± 1.26	59.50 ± 1.39	82.48 ± 1.84	81.15 ± 0.93	83.39 ± 0.12	80.37 ± 0.98	83.62 ± 1.93	80.11 ± 1.60	80.74 ± 2.20	**86.57 ± 0.10**
Kappa × 100	78.34 ± 0.44	73.02 ± 2.16	69.21 ± 1.91	78.66 ± 1.53	74.59 ± 1.30	77.66 ± 0.20	78.50 ± 1.07	79.03 ± 1.21	79.24 ± 1.03	78.26 ± 0.94	**80.69 ± 0.14**

network (EndNet) [63], CCR [49], global–local transformer (GLT) [41], hierarchical CNN and transformer (HCT) [53], HSI-LiDAR data based on Mamba (HLMamba) [47], mutual-guidance mechanism network (MGMNet) [64], multiscale cross-domain fusion network (MCFNet) [65], spatial–spectral–structural feature fusion network (S3F2Net) [34], and dual-feature attention hybrid GCN Mamba network (DAHGMN) [66]. All comparative methods are evaluated under the same data partitioning and evaluation metrics to ensure the fairness and comparability of the experimental results.

1) Quantitative Results and Analysis: Tables II–IV present a comprehensive comparison of classification accuracies among various methods on the Houston, MUUFL, and Trento datasets, where the best results are highlighted in bold, and the second-best results are underlined. The proposed method S2MSMamba consistently achieves the optimal results in terms of OA, AA, and Kappa, which substantiates the effec-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:08:01 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

QUANTITATIVE RESULTS (ACC% ± STD%) ON THE TRENTO DATASET (BEST RESULTS DISPLAYED IN BOLD AND SECOND-BEST RESULTS UNDERLINED)

No.	CoupledCNN	EndNet	CCR	GLT	HCT	HLMamba	MGMNet	MCFNet	S3F2Net	DAHGMN	S2MSMamba
C1	98.99 ± 0.49	88.74 ± 1.02	94.46 ± 3.23	99.33 ± 0.56	98.87 ± 0.63	99.37 ± 0.52	99.36 ± 0.23	99.50 ± 0.35	99.79 ± 0.17	99.17 ± 0.60	**99.94 ± 0.02**
C2	95.59 ± 2.85	95.81 ± 0.95	97.08 ± 2.21	97.72 ± 1.89	91.84 ± 5.96	97.24 ± 2.13	98.00 ± 1.27	97.94 ± 2.39	97.23 ± 1.83	**98.18 ± 2.04**	98.14 ± 0.37
C3	97.80 ± 1.37	95.62 ± 0.38	94.77 ± 4.44	96.99 ± 4.20	98.26 ± 0.89	98.45 ± 1.07	97.82 ± 1.35	97.67 ± 1.54	97.78 ± 0.78	98.41 ± 1.00	**100.00 ± 0.00**
C4	99.94 ± 0.15	99.42 ± 0.08	99.99 ± 0.02	**100.00 ± 0.00**	100.00 ± 0.00	100.00 ± 0.00	99.99 ± 0.01	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
C5	100.00 ± 0.01	76.55 ± 1.15	99.57 ± 0.19	**100.00 ± 0.00**	97.52 ± 3.20	100.00 ± 0.01	99.98 ± 0.03	99.98 ± 0.05	100.00 ± 0.01	99.94 ± 0.12	99.97 ± 0.01
C6	96.97 ± 1.94	90.45 ± 0.57	97.27 ± 1.84	97.27 ± 1.98	94.03 ± 2.51	96.61 ± 2.58	96.95 ± 1.53	96.68 ± 2.15	97.66 ± 1.08	96.43 ± 1.84	**98.54 ± 0.21**
OA	99.07 ± 0.33	88.69 ± 0.40	98.40 ± 0.53	99.36 ± 0.21	97.55 ± 1.29	99.27 ± 0.22	99.36 ± 0.16	99.35 ± 0.17	99.43 ± 0.20	99.30 ± 0.30	**99.65 ± 0.03**
AA	98.21 ± 0.61	91.10 ± 0.33	97.19 ± 1.17	98.55 ± 0.83	96.75 ± 1.10	98.61 ± 0.47	98.69 ± 0.39	98.63 ± 0.33	98.74 ± 0.42	98.69 ± 0.52	**99.43 ± 0.05**
Kappa × 100	98.76 ± 0.44	85.17 ± 0.50	97.87 ± 0.70	99.15 ± 0.28	96.75 ± 1.70	99.03 ± 0.30	99.15 ± 0.21	99.13 ± 0.23	99.23 ± 0.27	99.06 ± 0.40	**99.53 ± 0.04**

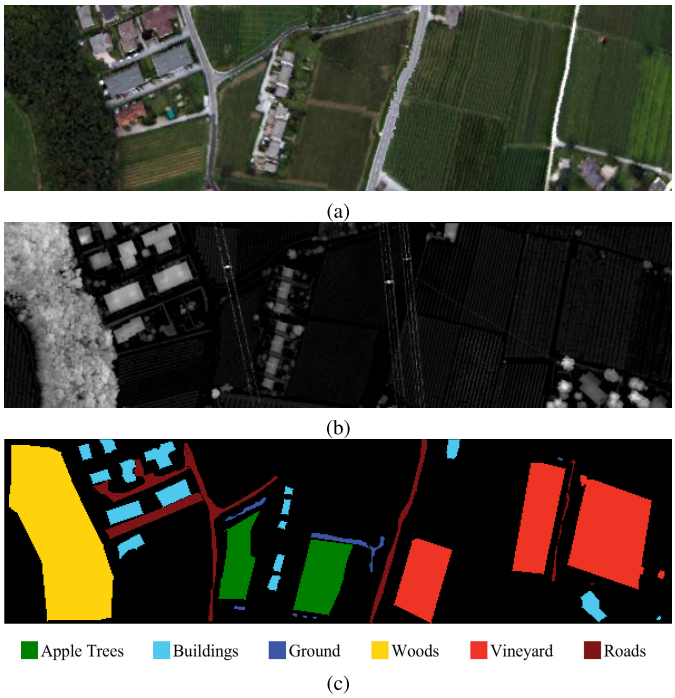


Fig. 4. Trento dataset. (a) RGB image, (b) grayscale image of LiDAR, and (c) ground-truth map.

tiveness of S2MSMamba in the multimodal HSI-LiDAR data classification tasks.

In the Houston dataset, as shown in Table II, quantitative results indicate that S2MSMamba yields an OA improvement of approximately 9.4% and 8.5% compared to autoencoder-based method EndNet and CNN-based method CCR, respectively, which effectively ameliorates the limitations of shallow networks in global context modeling. Furthermore, when compared with dual-branch methods, such as CNN-dual-branch method CoupledCNN, transformer-dual-branch method GLT, and CNN-transformer-dual-branch methods HCT and MGMNet, which have already attained high recognition rates in specific categories with prominent spectral features such as C14 and C15. S2MSMamba still achieves an OA increment of 1%–2% by leveraging the synergistic representation of global sequence dependencies and local spatial structures. The current advanced methods, such as HLMamba, MCFNet, S3F2Net, and DAHGMN, have already yielded a performance baseline above 95%. Furthermore, S2MSMamba effectively elevates the OA, AA, and Kappa to 96.09% and 96.71%, and 0.9574, which demonstrates that our Mamba-CNN-dual-branch structure of S2MSMamba possesses distinct advantages in the joint mining of long-range spectral dependencies and local spatial textures to effectively mitigate boundary confusion and scale variation among geographical features in urban environments.

In the MUUFL dataset, as shown in Table III, S2MSMamba achieves an OA gain of nearly 10% over EndNet and CCR. Furthermore, S2MSMamba outperforms the suboptimal method MCFNet by exhibiting improvements of 1.0% and 3.0% in OA and AA, respectively. Alongside these accuracy gains, S2MSMamba yields stable classification performance. S2MSMamba noticeably reduces the fluctuation of the standard deviations across all metrics, e.g., S2MSMamba decreases the AA standard deviation to 0.10%. This demonstrates that our proposed S2MSMamba can achieve outstanding steady classification results under limited training sample conditions. In extensive horizontal comparisons with CoupledCNN, GLT, HCT, HLMamba, MGMNet, S3F2Net, and DAHGMN, S2MSMamba generally delivers an OA approximately 2% higher and reaches a maximum AA increment of 6%. It can be seen that S2MSMamba can efficiently improve the classification performance by utilizing the adaptive decision-level fusion strategy to dynamically adjust multimodal weights. Furthermore, S2MSMamba effectively balances strong and weak modalities to prevent the former from overwhelming the latter, thereby achieving more robust feature discrimination in complex land-cover scenarios.

In the Trento dataset, as shown in Table IV, S2MSMamba demonstrates a significant OA improvement of approximately 10.96% compared to the autoencoder-based method EndNet, CNN-based method CCR and CoupledCNN, and CNN-transformer-based method HCT. Furthermore, when compared with current competitive multimodal methods, such as GLT, HLMamba, MGMNet, MCFNet, S3F2Net, and DAHGMN, which have already yielded an outstanding performance baseline above 99%. Furthermore, S2MSMamba effectively elevates the OA, AA, and Kappa to 99.65% and 99.43%, and 0.9953, respectively, and reaches the best or second-best accuracy in five out of the six land-cover categories, which establishes optimal performance across all global evaluation metrics. S2MSMamba successfully reduces the OA standard deviation to an unprecedented 0.03%, which explicitly confirms its outstanding algorithmic stability. S2MSMamba extracts highly robust spatial structural features by utilizing its multiscale hierarchical alignment mechanism, which demonstrates its excellent overall performance across all metrics.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:08:01 UTC from IEEE Xplore. Restrictions apply.

TABLE V
COMPUTATIONAL COMPLEXITY AND RUNNING TIME ON THE HOUSTON DATASET

Metrics	CoupledCNN	EndNet	CCR	GLT	HCT	HLMamba	MGMNet	MCFNet	S3F2Net	DAHGMN	S2MSMamba
FLOPs (G)	0.0023	0.0006	0.0004	0.1565	0.0079	0.0784	0.0691	0.0391	0.0575	0.0495	0.2201
Params (M)	0.1048	0.0920	0.0712	0.5608	0.4298	0.2214	3.4361	0.2523	0.2776	0.2889	0.6992
Training Time (s)	35.87	9.78	32.19	991.63	39.62	441.52	372.79	451.11	361.63	117.11	225.20
Test Time (s)	0.04	0.17	0.67	13.78	0.95	3.37	2.03	8.94	0.27	1.83	4.53

TABLE VI
COMPUTATIONAL COMPLEXITY AND RUNNING TIME ON THE MUUFL DATASET

Metrics	CoupledCNN	EndNet	CCR	GLT	HCT	HLMamba	MGMNet	MCFNet	S3F2Net	DAHGMN	S2MSMamba
FLOPs (G)	0.0023	0.0005	0.0004	0.0954	0.0080	0.0581	0.0695	0.0287	0.0142	0.0487	0.2039
Params (M)	0.1035	0.0891	0.0597	0.4009	0.4302	0.1983	3.4384	0.1915	0.3496	0.2837	0.6484
Training Time (s)	84.80	7.71	21.40	2764.75	39.18	1039.97	805.47	247.01	66.72	276.99	390.88
Test Time (s)	0.17	0.20	0.72	45.52	4.51	8.97	6.49	23.59	0.08	5.81	13.69

TABLE VII
COMPUTATIONAL COMPLEXITY AND RUNNING TIME ON THE TRENTO DATASET

Metrics	CoupledCNN	EndNet	CCR	GLT	HCT	HLMamba	MGMNet	MCFNet	S3F2Net	DAHGMN	S2MSMamba
FLOPs (G)	0.0023	0.0005	0.0004	0.0938	0.0079	0.0575	0.0691	0.0285	0.0109	0.0486	0.2035
Params (M)	0.1013	0.0887	0.0593	0.3963	0.4293	0.1972	3.4335	0.1898	0.2732	0.2828	0.6468
Training Time (s)	44.08	4.69	12.57	1496.95	14.57	518.07	453.91	398.20	56.84	157.84	236.27
Test Time (s)	0.08	0.18	0.43	25.14	1.85	4.09	3.60	13.35	0.09	3.36	7.75

The corresponding quantitative results are summarized in Tables V–VII. The experimental results show that S2MSMamba maintains competitive computational efficiency compared with most competing methods. Specifically, the model parameters (Params), training time, and test time are comparable to other existing methods. Although S2MSMamba requires higher floating-point operations (FLOPs) than some lightweight methods, the additional computational overhead mainly results from the integration of the Mamba module, the dual-branch feature extraction framework, and the dual-scale fusion strategy, which enhance the capability of modeling long-range dependencies and multiscale spatial information. Furthermore, S2MSMamba achieves better classification performance across all three benchmark datasets, which demonstrates a favorable balance between feature representation capability and computational efficiency.

2) Visualization Results and Analysis: Figs. 5–7 provide a visual comparison of the classification results obtained by various methods across the three datasets. Fig. 5 illustrates the visual classification results generated by various comparative methods on the Houston dataset. The proposed method S2MSMamba achieves the most visually coherent and structurally complete classification maps, which demonstrates that our proposed S2MSMamba can yield the optimal performance in regional consistency and boundary preservation. Specifically, autoencoder-based method EndNet and CNN-based method CCR generate classification maps plagued by severe salt-and-pepper noise and massive fragmented patches, while CoupledCNN marginally reduces the scattered noise but still fails to maintain structural integrity within large homogeneous regions, which exposes the limitations of shallow networks in spatial context modeling. The dual-branch methods, such as GLT, HCT, and MGMNet, significantly suppress the speckle noise, but they still exhibit noticeable boundary blurring and local class confusion when delineating complex urban structures and dense transitional zones. On the other hand, HLMamba, MCFNet, S3F2Net, and DAHGMN provide visual performance that is closer to the ground truth labels and yield highly competitive classification results. S2MSMamba preserves the sharp contours of small-scale targets and complex architectural boundaries while balancing noise suppression and detail preservation. This yields structurally complete and spatially realistic classification results in the complex urban environments.

Fig. 6 illustrates the visual classification results generated by various comparative methods on the MUUFL dataset. From the perspective of overall visual effects, the autoencoder-based method EndNet and the CNN-based method CCR generate classification maps plagued by noticeable salt-and-pepper noise and discontinuous geographical distributions in certain areas, which hinders the generation of stable and smooth classification results. The dual-branch methods, such as CoupledCNN, GLT, HCT, HLMamba, MGMNet, and DAHGMN, maintain the overall contours of geographical regions to a certain extent, but they still exhibit limitations in the recognition of complex categories and the preservation of spatial continuity, which triggers class confusion and boundary blurring phenomena in local areas. In contrast, current methods such as MCFNet and S3F2Net deliver significantly smoother classification results to effectively maintain the consistency and integrity of geographical regions. However, the detailed visual inspections of MCFNet and S3F2Net reveal minor misclassifications around subtle geographical details, which slightly degrade the precision of local contours. Compared to all comparative methods, the proposed S2MSMamba achieves the optimal visual classification performance by effectively suppressing the scattered noise and demonstrating superior capabilities in preserving geographical boundaries, regional

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:08:01 UTC from IEEE Xplore. Restrictions apply.

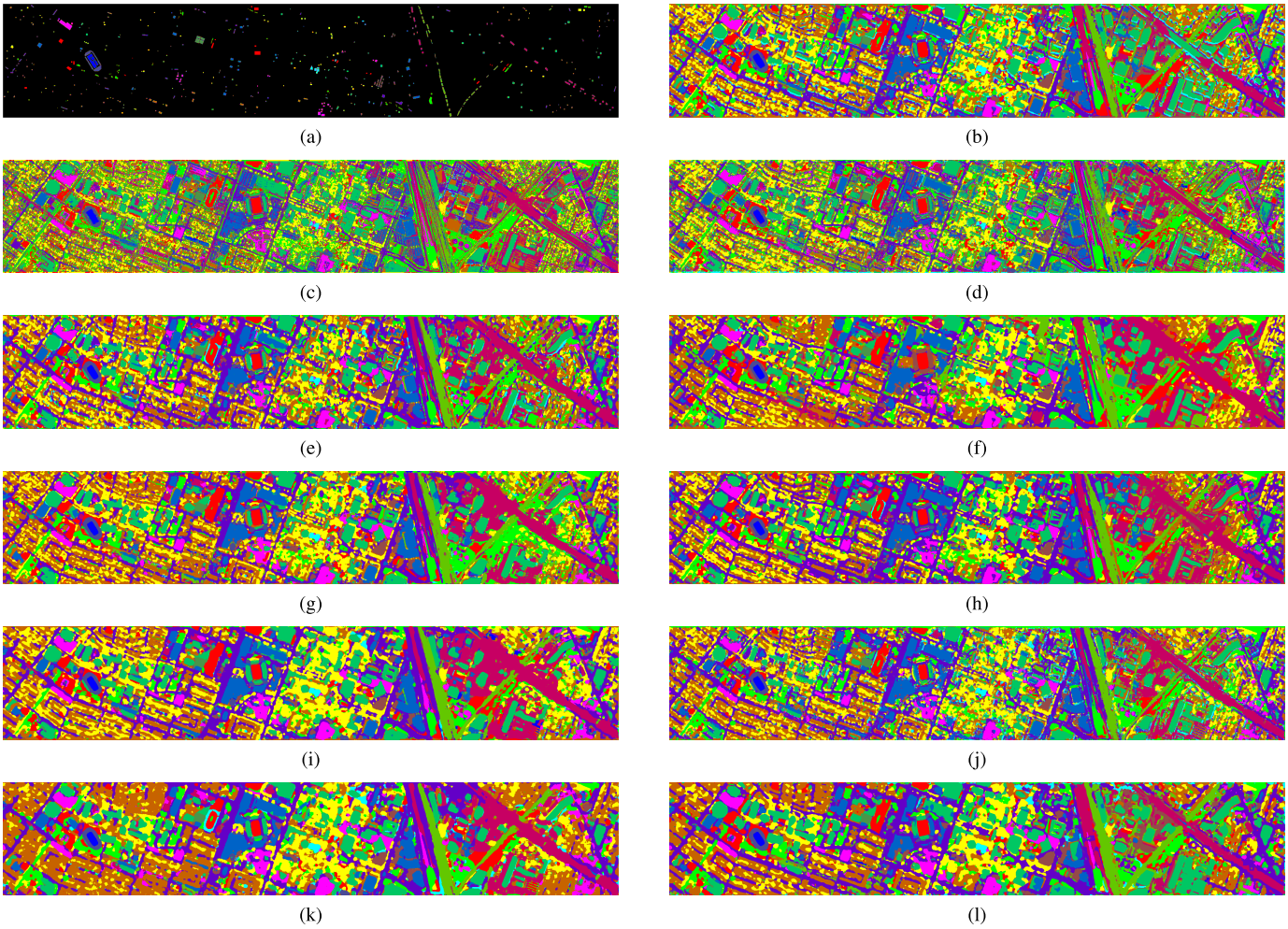


Fig. 5. Qualitative classification maps of different methods on the Houston dataset. (a) Ground-truth map, (b) CoupledCNN (93.82%), (c) EndNet (86.69%), (d) CCR (87.63%), (e) GLT (94.58%), (f) HCT (94.28%), (g) HLMamba (95.31%), (h) MGMNet (94.94%), (i) MCFNet (95.42%), (j) S3F2Net (94.71%), (k) DAHGMN (95.46%), and (l) S2MSMamba (96.09%).

integrity, and spatial continuity, which accurately delineates the spatial distribution characteristics among different land-cover categories to generate a clearer and more precise classification map.

Fig. 7 illustrates the visual classification results generated by various comparative methods on the Trento dataset. The autoencoder-based method EndNet and CNN-based method CCR generate classification results plagued by noticeable stripe-like noise and unstable class distributions in local areas. The CNN-transformer-based method HCT erroneously incorporates incorrect categories within certain large-scale homogeneous regions, which indicates distinct deficiencies during the spatial context modeling and feature aggregation processes. Along with the enhancement of feature representation capabilities, methods including CoupledCNN, GLT, HLMamba, MGMNet, MCFNet, S3F2Net, and DAHGMN significantly reduce noise interference to obtain smoother and more continuous classification results, but they still blur the geometric details at the intersections of different land covers. In contrast, the proposed S2MSMamba accurately recovers the boundary details and geometric structures of complex geographical features to effectively maintain the classification of small-scale targets and avoid class aliasing issues in edge regions, which ultimately generates the classification map closest to the authentic land-cover distribution.

TABLE VIII

MULTIMODAL DATA ABLATION RESULTS (%) ON HOUSTON, MUUFL, AND TRENTO DATASETS (OA ONLY)

Method	HSI Input	LiDAR Input	Houston	MUUFL	Trento
S2MSMamba-H	✓	×	93.84	82.34	98.06
S2MSMamba-L	×	✓	75.46	69.18	96.77
S2MSMamba	✓	✓	**96.09**	**85.04**	**99.65**

D. Ablation Study

To validate the necessity of multimodal data fusion and the complementary roles of the core components within the S2MSMamba framework, an ablation study is conducted from two perspectives: the effectiveness of multimodal data and the rationality of the network architecture. All quantitative results of the ablation variants are detailed in Tables VIII and IX.

1) Effectiveness of Multimodal Data: To validate the effectiveness of multimodal data integration, a multimodal data ablation study is conducted, which utilizes the complete

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:08:01 UTC from IEEE Xplore. Restrictions apply.

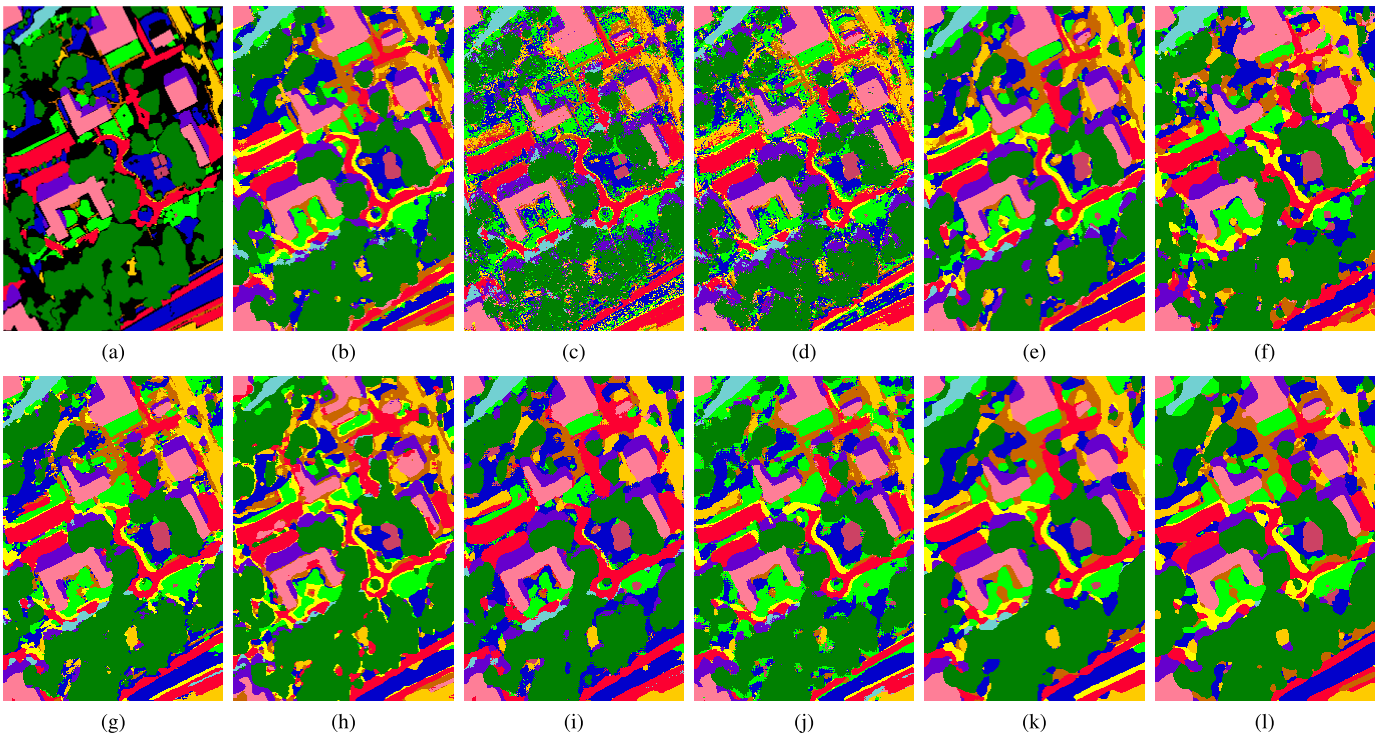


Fig. 6. Qualitative classification maps of different methods on the MUUFL dataset. (a) Ground-truth map, (b) CoupledCNN (83.18%), (c) EndNet (78.69%), (d) CCR (75.48%), (e) GLT (83.46%), (f) HCT (80.10%), (g) HLMamba (82.42%), (h) MGMNet (83.38%), (i) MCFNet (84.06%), (j) S3F2Net (84.00%), (k) DAHGMN (83.19%), and (l) S2MSMamba (85.04%).

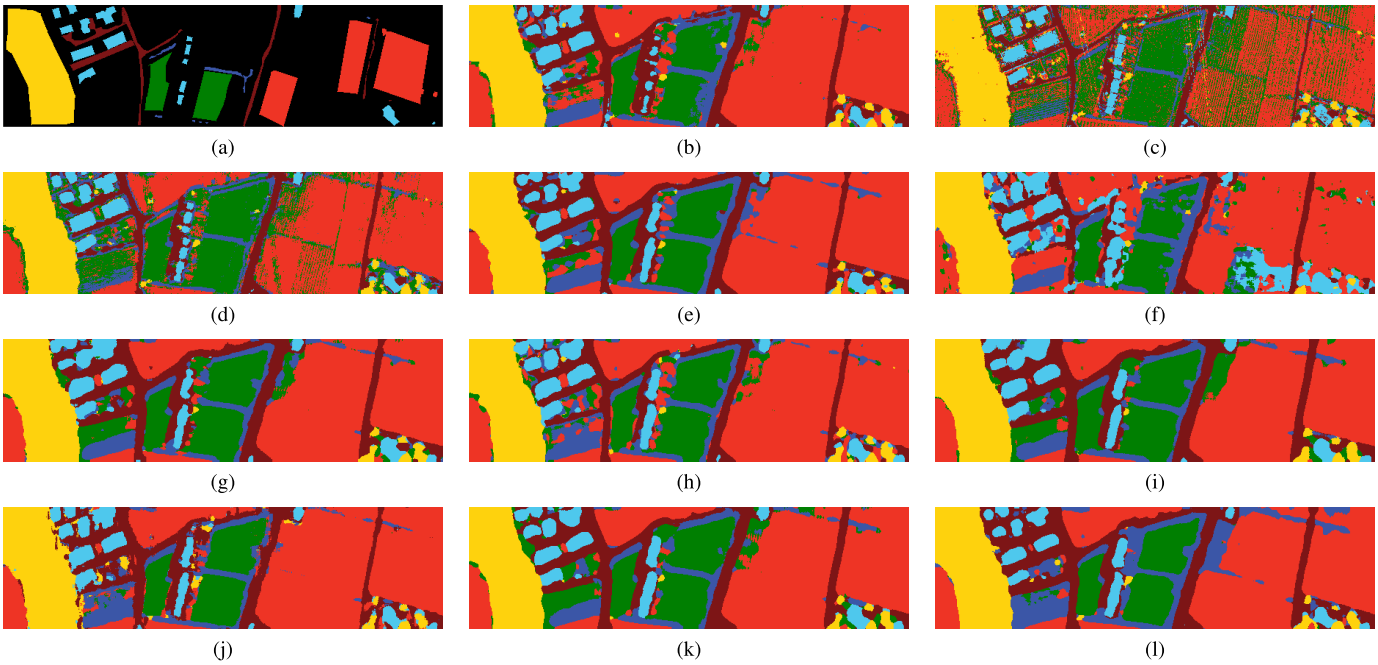


Fig. 7. Qualitative classification maps of different methods on the Trento dataset. (a) Ground-truth map, (b) CoupledCNN (99.07%), (c) EndNet (88.69%), (d) CCR (98.40%), (e) GLT (99.36%), (f) HCT (97.55%), (g) HLMamba (99.27%), (h) MGMNet (99.36%), (i) MCFNet (99.35%), (j) S3F2Net (99.43%), (k) DAHGMN (99.30%), and (l) S2MSMamba (99.65%).

proposed multimodal model as the baseline. Specifically, S2MSMamba-H removes the LiDAR data input and utilizes exclusively the HSI input for feature extraction. Meanwhile, S2MSMamba-L eliminates the HSI input and relies solely on the LiDAR data for feature extraction. Furthermore, unimodal variants of S2MSMamba-H and S2MSMamba-L inherently bypass the multimodal feature fusion and adaptive decision-level integration modules, which directly utilize their independent unimodal feature representations to generate the final classification result.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:08:01 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
CORE NETWORK COMPONENTS ABLATION STUDY RESULTS (%) ON HOUSTON, MUUFL, AND TRENTO DATASETS (OA ONLY)

Method	Multi-scale		Mamba	CNN	Feature	Decision	Houston	MUUFL	Trento
	8×8	16×16	Branch	Branch	Fusion	Fusion			
S2MSMamba-M_8	✓	×	✓	✓	✓	✓	94.00	82.57	98.89
S2MSMamba-M_{16}	×	✓	✓	✓	✓	✓	94.65	78.49	98.65
S2MSMamba-MA	✓	✓	×	✓	✓	✓	94.35	81.46	98.84
S2MSMamba-C	✓	✓	✓	×	✓	✓	94.89	79.14	98.69
S2MSMamba-F	✓	✓	✓	✓	×	✓	95.69	80.19	98.91
S2MSMamba-D	✓	✓	✓	✓	✓	×	95.06	80.44	99.13
S2MSMamba	✓	✓	✓	✓	✓	✓	**96.09**	**85.04**	**99.65**

Note: S2MSMamba-M_8 and S2MSMamba-M_{16} retain only the 8×8 and 16×16 scales, respectively; S2MSMamba-MA removes the Mamba branch; S2MSMamba-C eliminates the parallel CNN residual branch; S2MSMamba-F removes the deep feature fusion module; S2MSMamba-D removes the adaptive decision-level fusion mechanism.

As detailed in Table VIII, the proposed S2MSMamba achieves optimal classification performance through the joint integration of multimodal data, whereas the unimodal variants yield a reduction in classification performance due to the lack of complementary heterogeneous information. Specifically, S2MSMamba-H exhibits an OA reduction of approximately 3%. The performance degradation of S2MSMamba-H demonstrates that relying exclusively on HSI lacks the spatial elevation and structural attributes provided by LiDAR, which weakens the capability to differentiate spectrally similar ground objects that possess distinct physical profiles. Subsequently, S2MSMamba-L demonstrates an OA decrease of up to 20.63%, which constitutes the largest performance drop among all ablation variants. The substantial performance drop of S2MSMamba-L indicates that relying solely on LiDAR data completely discards the rich spectral discriminability provided by HSI, which severely weakens the ability to distinguish structurally similar but materially different land-cover categories. Overall, the quantitative results of the multimodal ablation study demonstrate that the joint utilization of HSI-LiDAR data can effectively exploit their complementary characteristics. Compared with S2MSMamba-H and S2MSMamba-L, the proposed multimodal framework S2MSMamba achieves better classification performance, which validates the effectiveness of the multimodal fusion strategy for HSI-LiDAR data classification.

2) Effectiveness of Core Network Components: To validate the necessity and complementary roles of the core components within the S2MSMamba framework, an ablation study is conducted, which utilizes the complete proposed model as the baseline. Specifically, S2MSMamba-M_8 removes the 16×16 scale input and retains only the 8×8 single-scale input for feature extraction, while S2MSMamba-M_{16} removes the 8×8 scale input and utilizes only the 16×16 single-scale input for feature extraction. In addition, S2MSMamba-MA eliminates the Mamba branch and relies solely on the parallel CNN branch for local spatial feature extraction. Meanwhile, S2MSMamba-C eliminates the parallel CNN residual branch and relies solely on the Mamba branch for the global extraction of sequence dependencies. Furthermore, S2MSMamba-F removes the deep feature fusion module and directly employs the weighted sum of unimodal predictions to generate the final classification result. Finally, S2MSMamba-D removes the classification probabilistic prediction of individual modalities and depends on the fused output $\mathbf{y}_f$ to make the final classification.

As detailed in Table IX, the proposed S2MSMamba achieves optimal classification performance through the joint integration of the core components, whereas the ablation variants yield a reduction in classification performance due to the structural inability to handle complex multimodal data. Specifically, S2MSMamba-M_8 and S2MSMamba-M_{16} demonstrate an OA reduction of approximately 7%. S2MSMamba-M_{16} exhibits a significantly larger performance degradation than S2MSMamba-M_8 on the MUUFL dataset, whereas the two single-scale variants exhibit comparable performance degradation on the Houston and Trento datasets. The performance degradation of S2MSMamba-M_8 and S2MSMamba-M_{16} indicates that relying on a single-scale input constrains the extraction of spatial contextual information at different resolutions and weakens the ability to handle scale variations among heterogeneous land-cover categories. In addition, S2MSMamba-MA demonstrates an OA decrease of up to approximately 4% across the datasets. The decline indicates that the absence of the Mamba branch limits the capture of global long-range spatial–spectral dependencies, thereby restricting the ability of S2MSMamba-MA to effectively model broad contextual information and leading to less discriminative global feature representations. Subsequently, S2MSMamba-C demonstrates an OA decrease of approximately 6%. The substantial decline indicates that the absence of the CNN branch limits the extraction of local spatial information and boundary details, thereby reducing the ability of S2MSMamba-C to distinguish similar land-cover categories and resulting in degraded classification performance. Furthermore, the structural isolation of data streams in S2MSMamba-F precludes explicit feature-level fusion. Consequently, S2MSMamba-F cannot effectively integrate complementary information from HSI-LiDAR modalities, which limits the ability of S2MSMamba-F to learn discriminative multimodal fusion and results in an OA decrease of approximately 5%. Finally, the exclusion of the adaptive decision-level fusion mechanism in S2MSMamba-D causes an OA decrease of approximately 5%. Consequently, S2MSMamba-D fails to effectively exploit the complementary information provided by multimodal and modality-specific classification predictions, thereby limiting its ability to adaptively balance different prediction sources and reducing OA performance. Overall, the quantitative results of the ablation study demonstrate that the proposed complete S2MSMamba framework provides structural rationality and is also an effective method for HSI-LiDAR classification.

VI. CONCLUSION

In this article, S2MSMamba is proposed as a novel S2MSMamba for HSI-LiDAR data classification. To extract modality-specific features, S2MSMamba designs a joint spatial–spectral Mamba block for HSI and a customized spatial Mamba block for LiDAR data, which enables S2MSMamba to extract discriminative spatial and spectral features. Subsequently, S2MSMamba deploys a multiscale hierarchical alignment mechanism, which efficiently resolves spatial misalignment among diverse ground targets. Furthermore,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:08:01 UTC from IEEE Xplore. Restrictions apply.

S2MSMamba utilizes an adaptive decision-level fusion strategy to dynamically balance multimodal and unimodal features, which effectively prevents the suppression of weaker modalities to maximize overall classification reliability. Through HSI-LiDAR experiments conducted on Houston, MUUFL, and Trento datasets, S2MSMamba establishes highly competitive performance baselines.

Although the proposed S2MSMamba achieves competitive classification performance on multiple benchmark datasets, there remains considerable potential for further improvement in both multimodal data fusion and network architecture design in the next work. First, future research will focus on extending the current dual-modal framework toward a more comprehensive heterogeneous multimodal fusion architecture. The integration of HSI, LiDAR, multitemporal observations, synthetic aperture radar (SAR) imagery, and auxiliary text information is expected to construct richer representations of complex geographical scenes and better exploit multimodal complementarities. In addition, more multibranch architectures, such as the Transformer-based branch, besides the current Mamba–CNN framework, can be considered to construct a new classification framework to further improve the discriminative power and robustness.

REFERENCES

[1] J. C. White, N. C. Coops, M. A. Wulder, M. Vastaranta, T. Hilker, and P. Tompalski, "Remote sensing technologies for enhancing forest inventories: A review," *Can. J. Remote Sens.*, vol. 42, no. 5, pp. 619–641, Sep. 2016.

[2] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "Deep learning classifiers for hyperspectral imaging: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, no. 1, pp. 279–317, Dec. 2019.

[3] B. Lu, P. Dao, J. Liu, Y. He, and J. Shang, "Recent advances of hyperspectral imaging technology and applications in agriculture," *Remote Sens.*, vol. 12, no. 16, p. 2659, Aug. 2020.

[4] W. Song, Z. Gao, R. Dian, P. Ghamisi, Y. Zhang, and J. A. Benediktsson, "Asymmetric hash code learning for remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jan. 2022, Art. no. 5617514.

[5] Z. Gao et al., "Joint learning of semantic segmentation and height estimation for remote sensing image leveraging contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, Jul. 2023, Art. no. 5614015.

[6] Q. Zhu et al., "A spectral–spatial-dependent global learning framework for insufficient and imbalanced hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 52, no. 11, pp. 11709–11723, Nov. 2022.

[7] Y. Ding et al., "Adaptive homophily clustering: Structure homophily graph learning with adaptive filter for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, Apr. 2025, Art. no. 5510113.

[8] Y. Ding et al., "SLCGC: A lightweight self-supervised low-pass contrastive graph clustering network for hyperspectral images," *IEEE Trans. Multimedia*, vol. 27, no. 1, pp. 8251–8262, Sep. 2025.

[9] Y. Ding et al., "Unsupervised self-correlated learning smoothy enhanced locality preserving graph convolution embedding clustering for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Aug. 2022, Art. no. 5536716.

[10] F. Liang et al., "DSGC: Dynamic sparse graph constrained autoencoder for hyperspectral image clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, Jun. 2026, Art. no. 5519012.

[11] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[12] W. Song, S. Li, L. Fang, and T. Lu, "Hyperspectral image classification with deep feature fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3173–3184, Jun. 2018.

[13] Y. Zhang, P. Duan, L. Liang, X. Kang, J. Li, and A. Plaza, "PFS3F: Probabilistic fusion of superpixel-wise and semantic-aware structural features for hyperspectral image classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 9, pp. 8723–8737, Sep. 2025.

[14] L. Liang, Y. Zhang, S. Zhang, J. Li, A. Plaza, and X. Kang, "Fast hyperspectral image classification combining transformers and SimAM-based CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, Aug. 2023, Art. no. 5522219.

[15] R. A. Borsoi et al., "Spectral variability in hyperspectral data unmixing: A comprehensive review," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 223–270, Dec. 2021.

[16] B. Zhao, M. O. Ulfarsson, J. R. Sveinsson, and J. Chanussot, "Hyperspectral image denoising using spectral–spatial transform-based sparse and low-rank representations," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jan. 2022, Art. no. 5522125.

[17] H. Zhang et al., "Hyperspectral image mixed noise removal based on sparsity constraint and low-rank approximation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, no. 12, pp. 17694–17714, May 2026.

[18] L. Fang, Z. Liu, and W. Song, "Deep hashing neural networks for hyperspectral image feature extraction," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 9, pp. 1412–1416, Sep. 2019.

[19] Z. Wang and M. Menenti, "Challenges and opportunities in LiDAR remote sensing," *Frontiers Remote Sens.*, vol. 2, Mar. 2021, Art. no. 641723.

[20] J. Jung, E. Pasolli, S. Prasad, J. C. Tilton, and M. M. Crawford, "A framework for land cover classification using discrete return LiDAR data: Adopting pseudo-waveform and hierarchical segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 2, pp. 491–502, Feb. 2014.

[21] J. Wang, W. Song, H. Chen, J. Ren, and H. Zhao, "FusDreamer: Label-efficient remote sensing world model for multimodal data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, Mar. 2025, Art. no. 5702314.

[22] D. Hong, N. Yokoya, G.-S. Xia, J. Chanussot, and X. X. Zhu, "X-ModalNet: A semi-supervised deep cross-modal network for classification of remote sensing data," *ISPRS J. Photogramm. Remote Sens.*, vol. 167, no. 1, pp. 12–23, Sep. 2020.

[23] M. Dalponte, L. Bruzzone, and D. Gianelle, "Fusion of hyperspectral and LiDAR remote sensing data for classification of complex forest areas," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 5, pp. 1416–1427, May 2008.

[24] M. Khodadadzadeh, J. Li, S. Prasad, and A. Plaza, "Fusion of hyperspectral and LiDAR remote sensing data using multiple feature learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2971–2983, Jun. 2015.

[25] M. Dalla Mura, J. A. Benediktsson, B. Waske, and L. Bruzzone, "Morphological attribute profiles for the analysis of very high resolution images," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 10, pp. 3747–3762, Oct. 2010.

[26] M. Pedergnana, P. R. Marpu, M. Dalla Mura, J. A. Benediktsson, and L. Bruzzone, "Classification of remote sensing optical and LiDAR data using extended attribute profiles," *IEEE J. Sel. Topics Signal Process.*, vol. 6, no. 7, pp. 856–865, Nov. 2012.

[27] J. Xia, N. Yokoya, and A. Iwasaki, "Fusion of hyperspectral and LiDAR data with a novel ensemble classifier," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 6, pp. 957–961, Jun. 2018.

[28] Y. Bengio, A. Courville, and P. Vincent, "Representation learning: A review and new perspectives," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 8, pp. 1798–1828, Aug. 2013.

[29] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, May 2015.

[30] S. Fang, K. Li, and Z. Li, "S^2ENet: Spatial–spectral cross-modal enhancement network for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, Jan. 2022.

[31] M. Zhang, W. Li, Q. Du, L. Gao, and B. Zhang, "Feature extraction for classification of hyperspectral and LiDAR data using patch-to-patch CNN," *IEEE Trans. Cybern.*, vol. 50, no. 1, pp. 100–111, Jan. 2020.

[32] Y. Zhang et al., "Multimodal transformer network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, Jun. 2023, Art. no. 5514317.

[33] K. Ni, D. Wang, Z. Zheng, and P. Wang, "MHST: Multiscale head selection transformer for hyperspectral and LiDAR classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, no. 1, pp. 5470–5483, Feb. 2024.

[34] X. Wang, L. Song, Y. Feng, and J. Zhu, "S3F2Net: Spatial–spectral-structural feature fusion network for hyperspectral image and LiDAR data classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 5, pp. 4801–4815, May 2025.

[35] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:08:01 UTC from IEEE Xplore. Restrictions apply.

[36] M. Zhang, W. Li, R. Tao, H. Li, and Q. Du, "Information fusion for classification of hyperspectral and LiDAR data using IP-CNN," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, no. 1, pp. 1–12, Jul. 2022.
[37] W. Song, Y. Dai, Z. Gao, L. Fang, and Y. Zhang, "Hashing-based deep metric learning for the classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, Oct. 2023, Art. no. 5704513.
[38] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, no. 4, pp. 3095–3110, Apr. 2022.
[39] Y. Zhang, Y. Peng, B. Tu, and Y. Liu, "Local information interaction transformer for hyperspectral and LiDAR data classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, no. 3, pp. 1130–1143, Jan. 2023.
[40] X. Zhao et al., "Fractional Fourier image transformer for multimodal remote sensing data classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 2, pp. 2314–2326, Feb. 2024.
[41] K. Ding, T. Lu, W. Fu, S. Li, and F. Ma, "Global–local transformer network for HSI and LiDAR data joint classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Oct. 2022, Art. no. 5541213.
[42] S. K. Roy, A. Deria, D. Hong, B. Rasti, A. Plaza, and J. Chanussot, "Multimodal fusion transformer for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, Jun. 2023, Art. no. 5515620.
[43] L. Wang and X. Wang, "Dual-coupled CNN-GCN-based classification for hyperspectral and LiDAR data," *Sensors*, vol. 22, no. 15, p. 5735, Jul. 2022.
[44] S. S. Rangapuram et al., "Deep state space models for time series forecasting," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 7796–7805.
[45] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," May 2023, *arXiv:2312.00752*.
[46] L. Huang, Y. Chen, and X. He, "Spectral–spatial mamba for hyperspectral image classification," *Remote Sens.*, vol. 16, no. 13, p. 2449, Jul. 2024.
[47] D. Liao, Q. Wang, T. Lai, and H. Huang, "Joint classification of hyperspectral and LiDAR data based on mamba," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, Sep. 2024, Art. no. 5530915.
[48] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.
[49] X. Wu, D. Hong, and J. Chanussot, "Convolutional neural networks for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–10, Nov. 2022.
[50] W. Song, Z. Gao, and Y. Zhang, "Discriminative feature extraction and fusion for classification of hyperspectral and LiDAR data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 2271–2274.
[51] X. Zhao et al., "Joint classification of hyperspectral and LiDAR data using hierarchical random walk and deep CNN architecture," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7355–7370, Oct. 2020.
[52] B. Yang, X. Wang, Y. Xing, C. Cheng, W. Jiang, and Q. Feng, "Modality fusion vision transformer for hyperspectral and LiDAR data collaborative classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, no. 2, pp. 17052–17065, Jun. 2024.
[53] G. Zhao, Q. Ye, L. Sun, Z. Wu, C. Pan, and B. Jeon, "Joint classification of hyperspectral and LiDAR data using a hierarchical CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, Jan. 2023, Art. no. 5500716.
[54] L. Sun, X. Wang, Y. Zheng, Z. Wu, and L. Fu, "Multiscale 3-D–2-D mixed CNN and lightweight attention-free transformer for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, Feb. 2024, Art. no. 2100116.
[55] Y. He, B. Tu, B. Liu, J. Li, and A. Plaza, "3DSS-mamba: 3D-spectral–spatial mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, Oct. 2024, Art. no. 5534216.
[56] C. Wang, J. Huang, M. Lv, H. Du, Y. Wu, and R. Qin, "A local enhanced mamba network for hyperspectral image classification," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 133, Sep. 2024, Art. no. 104092.
[57] G. Wang et al., "S2Mamba: A spatial–spectral state space model for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–13, Jan. 2025.
[58] M. Ahmad, M. Usama, M. Mazzara, and S. Distefano, "WaveMamba: Spatial–spectral wavelet mamba for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, Jan. 2025.
[59] J. Sheng, J. Zhou, J. Wang, P. Ye, and J. Fan, "DualMamba: A lightweight spectral–spatial mamba-convolution network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, Jan. 2025, Art. no. 5501415.
[60] A. Yang, M. Li, Y. Ding, L. Fang, Y. Cai, and Y. He, "GraphMamba: An efficient graph structure learning vision mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, Nov. 2024, Art. no. 5537414.
[61] F. Gao, X. Jin, X. Zhou, J. Dong, and Q. Du, "MSFMamba: Multiscale feature fusion state space model for multisource remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, Jan. 2025, Art. no. 5504116.
[62] M. Magnusson, J. Sigurdsson, S. E. Armansson, M. O. Ulfarsson, H. Deborah, and J. R. Sveinsson, "Creating RGB images from hyperspectral images using a color matching function," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Sep. 2020, pp. 2045–2048.
[63] D. Hong, L. Gao, R. Hang, B. Zhang, and J. Chanussot, "Deep encoder–decoder networks for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, Jan. 2022.
[64] W. Zhang, W. Song, J. Wang, and W. Gao, "MGMNet: Mutual-guidance mechanism for joint classification of multisource remote sensing data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, no. 1, pp. 1085–1097, Jan. 2025.
[65] Q. Song et al., "MCFNet: Multiscale cross-domain fusion network for HSI and LiDAR data joint classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, May 2025, Art. no. 4703912.
[66] Z. Xie, L. Lv, H. Gao, S. Xu, and H. Xie, "Dual-feature attention hybrid GCN mamba network for joint hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, Sep. 2025, Art. no. 5406514.



Bin Zhao (Member, IEEE) received the Ph.D. degree in electrical and computer engineering from the University of Iceland, Reykjavik, Iceland, in 2021.

From 2018 to 2023, he was a Scientist with the Faculty of Electrical and Computer Engineering, University of Iceland, under the supervision of Prof. Jon Atli Benediktsson, Prof. Magnus O. Ulfarsson, and Prof. Jocelyn Chanussot. He is currently a Professor and the Head of the Hyperspectral Remote Sensing Laboratory, College of Information Science and Engineering, Shandong Agricultural University, Tai'an, China. He received the Doctoral Grant of the University of Iceland Research Fund and the Icelandic Government Research Fund twice in 2018 and 2021, respectively. His research interests include hyperspectral image denoising, super-resolution, feature extraction, classification, and agricultural remote sensing.

Dr. Zhao was a recipient of the 2021 Chinese Government Award for the Outstanding Self-Financed Students Abroad. He is the first Chinese student in Iceland to receive the award. From all over the world, 650 young talents received the award from the China Scholarship Council. He has served as a Co-Chair for the IEEE International Geoscience and Remote Sensing Symposium in 2021. He is a Guest Editor of *Remote Sensing*. He is also the Leader of the Youth Innovation Team of Shandong Province of China.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:08:01 UTC from IEEE Xplore. Restrictions apply.



Wenjie Cao is currently pursuing the Master of Engineering degree from the College of Information Science and Engineering, Shandong Agricultural University, Tai'an, China.

Her research interests include hyperspectral image classification and multisource fusion classification.



Lin Han received the M.Sc. degree in geomatics engineering from China University of Geosciences (Wuhan), Wuhan, China, in 2016.

He is currently a Senior Engineer with the No.5 Geological Brigade of Shandong Provincial Bureau of Geology and Mineral Resources (No. 5 Institute of Geology and Mineral Exploration of Shandong Province), Tai'an, China. He also serves as Deputy Director of Shandong Provincial Laboratory of Big Data Development and Innovation for Geological Resources and Environment, and Deputy Director of Tai'an Data and Application Center for the High-Resolution Earth Observation System. His research interests include hyperspectral image processing, agricultural remote sensing, and geological remote sensing.



Weiwei Song received the Ph.D. degree in control science and engineering from Hunan University, Changsha, China, in 2021.

From November 2018 to November 2019, he was a Visiting Ph.D. Student under the supervision of Prof. Jon Atli Benediktsson with the Department of Electrical and Computer Engineering, University of Iceland, Reykjavik, Iceland, supported by the China Scholarship Council. From May 2021 to May 2024, he was a Post-Doctoral Researcher with Pengcheng Laboratory, Shenzhen, China. He is currently an Assistant Researcher with Pengcheng Laboratory. His research interests include deep learning, machine learning, and their applications in remote sensing.

Dr. Song was a recipient of the IEEE Geoscience and Remote Sensing Society 2024 Highest Impact Paper Award. He is an active Reviewer of several journals, such as the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, and IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT.



Wenjuan Zhang (Member, IEEE) received the B.S. degree in land resource management from Wuhan University, Wuhan, China, in 2003, and the Ph.D. degree in remote sensing from the Institute of Remote Sensing Applications, Chinese Academy of Sciences (CAS), Beijing, China, in 2008.

She is currently a Professor with the Airborne Remote Sensing Center, Aerospace Information Research Institute, CAS. Her research interests include remote sensing image quality improvement, image simulation, and payload characterize analysis.

Dr. Zhang received the Outstanding Science and Technology Achievement Prize of CAS in 2016 and the Second Prize of the State Scientific and Technological Progress Award in 2018.



Lianru Gao (Senior Member, IEEE) received the B.S. degree from Tsinghua University, Beijing, China, in 2002, and the Ph.D. degree from the Institute of Remote Sensing Applications, Chinese Academy of Sciences (CAS), Beijing, in 2007.

He is currently a Professor with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, CAS. He has published more than 300 peer-reviewed articles and there are more than 190 journal articles included in the Science Citation Index. He obtained 30 national invention patents in China. He was supported by the National Science Foundation as a Distinguished Young Scholar of China in 2023. His research interests include hyperspectral image processing and information extraction.

Dr. Gao is a fellow of the Institution of Engineering and Technology. He received the Outstanding Science and Technology Achievement Prize of CAS in 2016, the Second Prize of the State Scientific and Technological Progress Award in 2018, the IEEE Geoscience and Remote Sensing Society 2025 Highest Impact Paper Award, and the 2021 Outstanding Paper Award from the IEEE Workshop on Hyperspectral Image Processing: Evolution in Remote Sensing. He is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS).



Jakob Sigurdsson (Member, IEEE) received the M.S. degrees and the Ph.D. degree from the University of Iceland, in 2011 and 2015, respectively.

He is currently a Professor with the Department of Electrical and Computer Engineering, University of Iceland. His research interests include statistical signal- and image processing and remote sensing.



Magnus O. Ulfarsson (Senior Member, IEEE) received the B.S. and M.S. degrees from the University of Iceland, Reykjavik, Iceland, in 2002, and the Ph.D. degree from the University of Michigan, Ann Arbor, MI, USA, in 2007.

In 2007, he joined the University of Iceland, where he is currently a Professor and the Chair of the Faculty of Electrical and Computer Engineering. Since 2013, he has been with deCODE Genetics, Reykjavik. His research interests include statistical signal processing, image processing, machine learning, remote sensing, medical imaging, and genomics.

Dr. Ulfarsson has served as an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.



Jón Atli Benediktsson (Life Fellow, IEEE) received the Cand.Sci. degree in electrical engineering from the University of Iceland, Reykjavik, in 1984, and the M.S.E.E. and Ph.D. degrees in electrical engineering from Purdue University, West Lafayette, IN, USA, in 1987 and 1990, respectively.

Since 1991, he has been with the University of Iceland, where he has been Professor of Electrical and Computer Engineering since 1996. He served as a President and Rector of the University of Iceland from 2015 to 2025 and as a President of Aurora Universities from 2020 to 2024. He has been named to the Clarivate Highly Cited Researchers list since 2018. His research interests include remote sensing, biomedical signal analysis, pattern recognition, image processing, and signal processing, with extensive publications in these areas.

Prof. Benediktsson was the 2011-2012 IEEE Geoscience and Remote Sensing Society (GRSS) President and the 2003-2008 Editor-in-Chief of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. Currently, he is Editor-in-Chief of IEEE PRESS and Senior Editor for the PROCEEDINGS OF THE IEEE. He is an International Member of the National Academy of Engineering, a Fellow of SPIE, and a member of Academia Europaea, Sigma Xi, and Tau Beta Pi. He has received the IEEE Third Millennium Medal (2000), the IEEE GRSS Outstanding Service Award (2007), the GRSS David Landgrebe Award (2018), and the IEEE GRSS Education Award (2020). He was a co-recipient of the 2012 IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSINGPaper Award. In 2013 and 2024, he was a co-recipient of the IEEE GRSS Highest Impact Paper Award.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:08:01 UTC from IEEE Xplore. Restrictions apply.